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PROTECTION AND METERING INSTRUMENT TRANSFORMERS CT & PT SIZING CALCULATION

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TRANSMISSION ENGINEERING STANDARD

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1.0 PURPOSE AND SCOPE

This engineering standard establishes the general guidelines; parameters and design criteria for sizing of Current Transformers (CTs) and Potential Transformers (PTs) intended to be used in the Transmission System of Saudi Electricity Company, Kingdom of Saudi Arabia.

This engineering standard includes the criteria for selection of various types of current transformers and potential transformers per applications of protection and measuring devices.

2.0 SYSTEM PARAMETERS

The system parameters for various system voltage levels shall be referred to the latest revision of SEC General Specification 01-TMSS-01, the General Requirements for all equipment/materials or as specified in the S.O.W/P.T.S.

3.0 NOMENCLATURE OF PROTECTION / MEASURING DEVICES AND NUMBERING SYSTEM

The protective devices usually initiated by the instrument transformers are listed below.

The numbering system to identify the protective/metering devices nomenclature as shown below shall be adopted on schematic and wiring diagrams as per IEEE C37-2 (2008) and affixed to the relay cover and relay carriage.

The complete device function numbers shall comprise of a number, possibly followed by a suffix letter(s) indicating the type of primary protection protected. This whole device function number may then, if appropriate, be followed by a suitable designation to a particular transformer or section of bus bar etc.

NAME OF THE DEVICE	DEVICE NUMBER
Time delay starting or closing relay	2
Checking or Interlocking Relay	3
Distance Relay (Phase to Phase Protection)	21
- Distance Protection Phase to Earth	21 G
- Distance Protection Zone 1	21-Z1
- Distance Protection Zone 2	21-Z2
- Distance Protection Zone 3	21-Z3
- Distance Protection Zone 4	21-Z4
Room Temperature High (Temperature Control Device)	23
Synchro Check Relay	25
- System Synchronizing Relay	25 S

NAME OF THE DEVICE	DEVICE NUMBER
Battery Faulty (Hi-V, Lo-V, Hi-Z)	27/59 B
Under Voltage Relay	27
Trip Circuit Supervision Relay	27 TC
Close Circuit Supervision Relay	27 CC
Transformer HV On-Load Isolator	29 T
Annunciator Relay for Local Indication & Remote Alarm	30
Directional Power Relay	32
Bearing Protective Device	38
Manual Transfer or Selector Switch	43
Smoke Detector (Atmospheric condition monitor)	45
Phases Not Together (Incomplete sequence relay)	48
Machine or Transformer Thermal Relay	49
Winding Temperature Alarm	49 WG
Winding Temperature Trip	49 T
Instantaneous Over-current (O/C) Non-directional	50
Circuit Breaker Fail Current Check	50 BF
Instantaneous E/F Non-directional	50 G
High Instantaneous Over-current (O/C) relay	50 H
Instantaneous Definite Minimum Time (IDMT) O/C Non-directional	51
Stuck Breaker Protection – 13.8 kV Phase and E/F	51 B/51 NB
Definite Time O/C Non-directional	51 DT
I.D.M.T. E/F Non-directional	51 G
Feeder O/C Non-directional	51 F
Feeder Earth Fault (E/F) Non-directional	51 NF
Feeder O/C Inst. / I.D.M.T.	50/51 F
Transformer H.V. O/C I.D.M.T	51 P
Transformer H.V. O/C Inst. / I.D.M.T	50/51 P
Transformer L.V. O/C	51 S
Transformer H.V. E/F	51 NP
Transformer L.V. E/F	51 NS
Standby E/F Protection	51 N
Two Stages, Standby E/F Protection Stage 1	51 N1
Two Stages, Standby E/F Protection Stage 2	51 N2

NAME OF THE DEVICE	DEVICE NUMBER
Neutral Current E/F Alarm	51 NA
A.C. Circuit Breaker	52
Circuit Breaker for Transformer 1	52 T1
Circuit Breaker for Transformer Secondary 1	52 S1
Circuit Breaker for Line 1	52 L1
Circuit breaker for Bus Coupler 1	52 BC1
Circuit breaker for Bus Section 1	52 BS1
Earth Switch (short-circuiting or grounding device)	57
Earth Switch for Transformer 1	57 T1
Earth Switch for Line 1	57 L1
Earth Switch for Bus Section 1	57 BC2A/2B
Earth Switch for Bus Coupler 1	57 BS2A/2B
Over voltage Relay	59
Neutral Voltage Displacement Relay	59 N
Time Delay for tripping (stopping or opening)	62
Circuit Breaker Fail Timer	62 BF
Air Pressure Switch of Circuit Breaker	63AC
Gas Pressure Switch of Circuit Breaker	63GA
Gas Pressure High in Bushing	63GB
Gas Pressure Switch of GIS	63GL
Oil Flow Relay	63QF
Oil Level Low- Transformer	63QL-1
Oil Level Low-Tap Changer	63QL-2
Tap Changer protective Device (Trip)	63QP
Low Oil Pressure	63QPL
Pressure Switch followed by a Suffix	63
Battery Earth Fault	64 B
Transformer H.V. Restricted Earth Fault (R.E.F)	64 NP
Transformer L.V. R.E.F	64 NS
Directional O/C	67
Directional E/F	67 N
Feeder Directional O/C	67 F

NAME OF THE DEVICE	DEVICE NUMBER
Feeder Directional E/F	67 NF
Transformer L.V. Directional O/C	67 S
Transformer L.V. Directional E/F	67 NS
Blocking Relay (Power Swing Out of Step)	68
Alarm Relay	74
Phase-Angle measuring relay	78
AC Reclosing Relay	79
Under Frequency Relay	81 UF
Ferro Resonance Detection Relay	81 FR
Inter trip Receive Relay	85 R
Inter trip Receive Follower Relay	85 RX
Carrier Equipment Failure	85 F
Inter trip Send	85 S
Trip Relay Electric/Hand Reset (Lockout Relay)	86
Busbar Protection Trip Relay	86 B
Connections Trip Relay	86 C
Feeder Protection Trip Relay	86 F
Feeder End Protection Trip Relay	86 FE
Reactor Protection Trip Relay	86 R
Transformer HV Trip Relay	86 T
Bus Bar Differential Protection	87 B
Bus Bar Protection Section ‘A’	87 B1
Bus Bar Protection Section ‘B’	87 B2
Bus Bar Protection Section ‘C’	87 B3
Bus Bar Protection Section ‘A’ Front	87 BA2
Bus Bar Protection Section ‘A’ Rear	87 BA1
Bus Bar Protection Section ‘B’ Front	87 BB2
Bus Bar Protection Section ‘B’ Rear	87 BB1
Connections Circulating Current Protection	87 C
Feeder End Protection	87 FE
Cable Differential Protection	87 PW
Transformer Overall Differential Protection	87 T
Transformer End Protection	87 TE

NAME OF THE DEVICE	DEVICE NUMBER
Electrically Operated Isolator (Line Switch)	89
Electrically Operated Isolator for Transformer 1	89 T1
Electrically Operated Isolator for Line 1	89 L1
Electrically Operated Isolator for Bus Coupler	89 BC2F/2R
Electrically Operated Isolator for Bus Section	89 BS2A/2B
Voltage Regulating relay	90
Trip Relay Self-Reset	94
Remote Emergency Trip Receive	94 R
Transformer LV Trip Relay	94 T
CT Supervision Relay	95
Buchholz Gas alarm	96-1
Buchholz Surge Trip	96-2
Burst Plate Breaking Device	96D
PT fuse failure relay	98
Line Drop Compensating Unit for Automatic Voltage Control Equipment	LDC
Hand Operated Earthing Switch	ES
Un-Stabilizing Relay	USB
Event Recorder / Fault Locator	ER/FL

4.0 DESCRIPTION OF PROTECTION RELAYS / DEVICES

4.1 Device Number 2 – Time delay starting or closing relay

A device that functions to give a desired amount of time delay before or after any point of operation in a switching sequence or protective relay system, except as specifically provided by device functions 48, 62, 79.

4.2 Device Number 3 – Checking or interlocking relay

A device that operates in response to the position of one or more devices or predetermined conditions in a piece of equipment or circuit, to allow an operating sequence to proceed, or to stop, or to provide a check of the position of these devices or conditions for any purpose.

4.3 Device Number 21 – Distance relay

A device that functions when the circuit admittance, impedance, or reactance increases or decreases beyond a predetermined value.

4.4 Device Number 23 – Temperature control device

A device that functions and controls the temperature of a machine or apparatus, or of any medium, when its temperature falls below or rises above a predetermined value.

4.5 Device Number 25 – Synchronizing or synchronism check relay
A synchronizing device produces an output that causes closure at zero-phase angle difference between two circuits. It may or may not include voltage and speed control. A synchronism-check relay permits the paralleling of two circuits that are within prescribed limits of voltage magnitude, phase angle, and frequency.

4.6 Device Number 26 – Apparatus thermal device
A device that functions when the temperature of the protected apparatus (other than the load-carrying windings and transformers as covered by device function number (49) or of a liquid or other medium exceeds a predetermined value; or when the temperature of the protected apparatus or of any medium decreases below a predetermined value.

4.7 Device Number 27 – Under voltage relay
A device that operates when its input voltage is less than a predetermined value.

4.8 Device Number 29 – Isolating contactor or switch
A device used expressly for disconnecting one circuit from another for the purpose of emergency operation, maintenance or test.

4.9 Device Number 30 – Annunciator relay
A non-automatically reset device that gives a number of separate visual indications upon the functioning of protective devices and that may also be arranged to perform a lock out function.

4.10 Device Number 32 – Directional power relay
A device that operates on a predetermined value of power flow in a given direction such as reverse power flow resulting from the motoring of a generator upon loss of its prime mover.

4.11 Device Number 38 – Bearing Protective Device
A device that functions on excessive bearing temperature or on other abnormal mechanical conditions associated with the bearing, such as undue wear, which may eventually result in excessive bearing temperature or failure.

4.12 Device Number 43 –Selector Switch
A manually operated device that transfers and control the potential circuits in order to modify the plan of operation of the associated equipment or of some of the associated devices.

4.13 Device Number 45 – Atmospheric condition monitor
A device that functions upon the occurrence of an abnormal atmospheric condition such as damaging fumes, explosive mixtures, smoke, or fire.

4.14 Device Number 48 – Incomplete sequence relay
A device that generally returns the equipment to the normal or off position and locks it out of the normal starting, operating, or stopping sequence is not properly completed within a predetermined time.

4.15 Device Number 49 – Machine or Transformer Thermal Relay
A device that functions when the temperature of a machine armature winding or other load carrying winding or element of a machine or power transformer exceeds a predetermined value.

4.16 Device Number 50 – Instantaneous over current relay
A device that operates with no intentional time delay when the current exceeds a preset value.

4.17 Device Number 51 – AC time over current relay
A device that functions when the AC input current exceeds a predetermined value, and in which the input current and operating time are inversely related through a substantial portion of the performance range.

4.18 Device Number 52 – AC circuit breaker
A device that is used to close and interrupt an AC power circuit under normal conditions or to interrupt this circuit under fault or emergency conditions.

4.19 Device Number 57 – Short circuiting or grounding device
A device that functions to short circuit or ground a circuit in response to automatic or manual means.

4.20 Device Number 59 – Over voltage relay
A device that operates when its input voltage exceeds a predetermined value.

4.21 Device Number 62 – Time delay stopping or opening relay
A device that imposes a time delay in conjunction with the device that initiates the shutdown, stopping or opening in an automatic sequence or protective relay system.

4.22 Device Number 63 – Pressure switch
A device that operates at a given pressure value or at a given rate of change of pressure.

4.23 Device Number 64 – Ground detector relay
A device that operates upon failure of insulation to ground of machine or other apparatus.

4.24 Device Number 67 – AC directional over current relay
A device that functions to desired value of AC current flowing in a predetermined direction.

4.25 Device Number 68 – Blocking or “out-of-step” relay
A device that initiates a pilot signal for blocking of tripping on external faults in a transmission line or in other apparatus under predetermined conditions, or co-operates with other devices to block tripping or reclosing on an out-of-step condition or on power swings.

4.26 Device Number 74 – Alarm relay
A device other than an Annunciator, as covered under device function 30, that is used to operate, or that operates in connection with, a visual or audible alarm.

4.27 Device Number 78 – Phase-angle measuring relay
A device that functions at a predetermined phase angle between two voltages, between two currents, or between voltage and current.

4.28 Device Number 79 – Auto Reclosing Relay
A device that controls the automatic reclosing and locking out of an AC circuit interrupter.

4.29 Device Number 81 –Frequency relay
A device that responds to the frequency of an electrical quantity, operating when the frequency or rate of change of frequency exceeds or is less than a predetermined value.

4.30 Device Number 85 – Carrier or pilot-wire relay
A device that is operated or restrained by a signal transmitted or received via any communications media used for relaying.

4.31 Device Number 86 – Lock out relay
A device that trips and maintains the associated equipment or devices inoperative until it is reset by an operator, either locally or remotely.

4.32 Device Number 87 – Differential protective relay
A device that operates on a percentage, phase angle, or other quantitative difference of two or more currents or other electrical quantities.

4.33 Device Number 89 – Electrically Operated Isolator
A device used as a disconnecting, line isolators, or isolating switch in an AC or DC power circuit. This device number is for switches that are electrically operated or has electrical accessories, such as an auxiliary switch, a magnetic lock, etc.

4.34 Device Number 90 – Regulating device
A device that functions to regulate a quantity or quantities, such as voltage, current, power, speed, frequency, temperature, and load, at a certain value or between certain (generally close) limits for machines, tie lines, or other apparatus.

4.35 Device Number 94 – Tripping or trip-free relay
A device that functions to trip a circuit breaker, contactor, or equipment; to permit immediate tripping by other devices; or to prevent immediate reclosing of a circuit interrupter if it should open automatically, even though its closing circuit is maintained closed.

4.36 Device Number 97 – 99 – Used for specific applications
These device numbers are used in individual specific installations if none of the functions assigned to the numbers from 1 through 96 are suitable.

5.0 INSTRUMENT CURRENT TRANSFORMERS

5.1 General

Current Transformer is a type of instrument transformer designed to provide a current in its secondary winding proportional to the alternating current flowing in its primary. It is commonly used in metering and protection relaying in the electrical power industry where it facilitates the safe measurement of large currents often in the presence of high voltages. The current transformer safely isolates measurement and control circuitry from the high voltages typically present on the circuit being measured.

Current transformers are used extensively for measuring current and monitoring the operation of the power grid. The CT is typically described by its current ratio from primary to secondary. Often, multiple CTs are installed as a "stack" for various uses (for example, protection devices and revenue metering may use separate CTs).

Shapes and sizes can vary depending on the end user or switchgear manufacture. Typical examples of low voltage single ratio metering current transformers are either ring type or cast resin type.

Depending on the requirement, there are two main standards to which current transformers are designed. IEC 60044-1 (BSEN 60044-1) & IEEE C57.13 (ANSI). The most common design of CT consists of a length of wire wrapped many times around a silicon steel ring passed over the circuit being measured. The CT's primary circuit therefore consists of a single 'turn' of conductor, with a secondary of many hundreds of turns.

The CT acts as a constant-current series device with an apparent power burden a fraction of that of the high voltage primary circuit. Hence the primary circuit is largely unaffected by the insertion of the CT.

Common secondaries are 1 or 5 amperes. The secondary winding can be single ratio or multi ratio, with five taps being common for multi ratio CTs.

For IEC (BSEN) typically, the secondary connection points are labeled as 1S1, 1S2, 2S1, 2S2 etc for multi cores CT's or S1, S2, etc for single core single ratio CT's, or in the ANSI/IEEE standard areas, X1...X5, Y1...Y5, and so on. The multi ratio CTs are typically used for current matching in current differential protective relaying applications.

For a three-stacked CT application, the secondary winding connection points are typically labeled X_n, Y_n, Z_n.

The burden in a CT metering circuit is essentially the amount of impedance (largely resistive) present. Typical burden ratings for IEC CTs are 1.5VA, 3VA, 5VA, 10VA, 15VA, 20VA, 30VA, 45VA & 60VA with ANSI/IEEE B-0.1, B-0.2, B-0.5, B-1.0, B-2.0 and B-4.0. This means a CT with a burden rating of B-0.2 can tolerate up to 0.2 Ω of impedance in the metering circuit before its output current is no longer a fixed ratio to the primary current. Items that contribute to the burden of a current measurement circuit are switch blocks meters and intermediate conductors. The most common source of excess burden in a current measurement circuit is the conductor

between the meter and the CT. Often, substation meters are located significant distances from the meter cabinets and the excessive length of small gauge conductor creates a large resistance. This problem can be solved by using CT with 1 ampere secondaries which will produce less voltage drop between a CT and its metering devices or by using CT Cable of larger Cross Sectional Area.

Burden class IEEE/ANSI (Also called Knee-point Voltage) is the voltage at which a CT becomes saturated. When a CT becomes saturated it can no longer transform current. An example of this rating would be C200, C800 etc. This means that the CT will saturate near 200 Volts or 800 Volts respectively.

5.2 Rating Factor

Rating factor is a factor by which the nominal full load current of a CT can be multiplied to determine its absolute maximum measurable primary current. Conversely, the minimum primary current a CT can accurately measure is "light load," or 10% of the nominal current (there are, however, special CTs designed to measure accurately currents as small as 2% of the nominal current). The rating factor of a CT is largely dependent upon ambient temperature. Most CTs have rating factors for 35 degrees Celsius and 55 degrees Celsius. It is important to be mindful of ambient temperatures and resultant rating factors when CTs are installed inside poorly ventilated mechanical rooms.

5.3 CT configuration

Physical CT configuration is another important factor in reliable CT accuracy. When conductors passing through a CT are not centered in the circular (or oval) void, slight inaccuracies may occur. It is important to center primary conductors as they pass through CTs to promote the greatest level of CT accuracy.

In power systems applications, many CT configurations bypass this limitation by either being custom designed to slip around the bushing of a high-voltage transformer or circuit breaker, which automatically centers the conductor inside the CT window, or by having the primary turn permanently fixed inside the CT housing and accessible only by means of external terminals.

5.4 Current Transformer Classes for Protection Devices

The voltage produced from the CT secondary windings is to be determined according to its class rating, turns ratio and up to what point is the voltage and current of the transformer core is magnetically saturated.

Through the basic ohm's law application, calculations of the size or class of CT needed for particular purpose can be determined.

Current Transformers for protection are classified according to its functional performance.

CT classes according to IEC 60044-1 protective current transformer accuracy class, Class P, are defined as 5P and 10P with the following limits per Table 1.0 below:

extracted from the standard. However, Class PX current transformer if specifically required in Data Schedule or in SOW/PTS shall be preferred over class 5P or 10P current transformer.

Table 1.0 Limits of Error for Protective Current Transformers

Accuracy class	Current error at rated primary current %	Phase displacement at rated primary current		Composite error at rated accuracy limit primary current %
		minutes	centi-radians	
5P	±1	±60	±1.8	5
10P	±3	-	-	10

5.5 Current Transformer Classes for Protection Devices as per IEEE C 57.13 the CT protection relaying accuracy class is determined by letter designation and secondary terminal voltage rating.

CT bushing type Class C and Class K shall be used for protection relaying purposes.

The classification number indicates the voltage that the current transformer will deliver to a standard burden at 20 times its normal secondary current without exceeding a 10% ratio error.

CTs shall be designated for example as C-100, C-200, etc. as a result of calculations considering all factors which may influence the rating of CT, such as burden of connected relay together with the lead resistance ohmic values, etc.

The CT relaying accuracy class voltage shall be chosen so that the CT secondary output shall be sufficient to ensure proper performance of the associated relays.

5.6 Current Transformer for measuring devices

The accuracy class of metering CT depends on the load currents. For measuring current transformers, the accuracy class is designated by the highest permissible percentage current error at rated current prescribed for the accuracy class concerned. Current Transformer for metering by IEC designation are to be identified by the code letter “M” preceded by the accuracy class and followed by the multiplier for the maximum primary current.

As a general guide, the following are the typical class of accuracy adopted in SEC system:

CT class 0.1 to 0.2 is generally adapted for precision measurements
CT class 0.5 is for high grade kilowatt hour meters.
CT class 3 to 5 is for approximate measurements.

The measurement current transformer shall have the same ratio and thermal current limit as the associated protection CTs in the circuit.

5.7 Current Transformer Sizing Formulae for Protection and Measuring Devices

5.7.1 Design Parameters

The value of resistance for CT winding and lead wires used for the terminal connections of the current transformers, relay and meters and auxiliary CTs is generally based on reference temperature of 20ºC.

A twenty (20%) percent margin on calculated knee-point voltage is to be considered for the CT not to be saturated.

A twenty-five (25%) percent margin on the calculated CT burden of the connected load is considered to give enough safety factors and prevent CT to be saturated under all condition.

5.7.2 CT Knee-point Voltage

The CT knee-point voltage is the point on the excitation curve where the tangent is 45° to the abscissa. This definition is for non-gapped CTs. When the CT has a gapped core, the knee-point voltage is the point where the tangent to the curve makes an angle of 30° with the abscissa.

One of the factors which influence the class of current transformer to be adopted is its knee-point voltage (V_{kp}). It is the secondary voltage to be calculated and determined at which an increase of 10% produces an increase in magnetizing current of 50%. It is also the secondary voltage above which the current transformer will become saturated.

In practical, CT saturation can be avoided by sizing the CT to have a knee-point voltage above that required for maximum expected fault current and CT secondary burden.

Computation for the CT burden shall be based on the actual connected load of the CT which includes the sum of connecting lead wires resistance, impedance of auxiliary CTs (if any) and impedance of connected relays and meters and CT internal burden due to CT secondary winding resistance (Rct).

5.7.3 Resistance of CT Lead Wires

Below is the formula for calculating the total resistance of lead wires.

$R_L = R_{20} \cdot L \cdot [1 + \acute{\alpha} \cdot (T - 20)]$, wherein

R_L is the lead wire resistance (two way length)
L is the total length of CT lead wires as per measurement
R_{20} is the CT lead resistance per unit length at 20ºC, per IEC 60228 and Table 2.0 below.

ά is the Temperature Coefficient of Resistance of copper conductor at inferred absolute zero of temperature (ά ≈ 1/234.5)
T is the final temperature of the wires (T=75 ºC)

Table 2.0- Resistance of Stranded Copper Conductor for Single–core and Multi-core Cables.

Nominal cross-sectional area mm^2	Minimum number of wires in the conductor			Maximum resistance at 20ºC of Annealed copper conductor (Plain wires) Ω/km
	Circular	Circular compacted	Shaped	
	Cu	Cu	Cu	
0.5	7	-	-	36.0
0.75	7	-	-	24.5
1.0	7	-	-	18.1
1.5	7	6	-	12.1
2.5	7	6	-	7.41
4	7	6	-	4.61
6	7	6	-	3.08
10	7	6	-	1.83
16	7	6	-	1.15
25	7	6	6	0.727
35	7	6	6	0.524
50	19	6	6	0.387
70	19	12	12	0.268
95	19	15	15	0.193
120	37	18	18	0.153
150	37	18	18	0.124
185	37	30	30	0.0991
240	37	34	34	0.0754
300	61	34	34	0.0601
400	61	53	53	0.0470
500	61	53	53	0.0366
630	91	53	53	0.0283

5.7.4 Various Formulae for CT Knee-point Voltage Calculation as per Protection Requirements.

The following formula shall be used as a guide in computing the correct Knee point voltage and sizing of the current transformer based on the protection scheme requirements:

5.7.4.1 High Impedance Differential Relay

$V_{KP} \geq 2 \times I_{K2} \times (R_{CT} + R_L)$,

Wherein;
V_{KP} is the knee-point voltage of the current transformer,
I_{K2} is the maximum through fault current related to secondary side of CT,
$I_{K2} = I_K/n$
Wherein;

I_K is the maximum through fault current related to the primary side of CT.
n = CT ratio

R_{CT} is the total secondary resistance of current transformer per protection requirement,
R_L is the total (two way length) CT lead resistance

5.7.4.2 Inverse Time Function connection

$$V_{KP} \geq 20 \times I_{set} \times (R_{CT} + R_L + R_R),$$

Wherein;
I_{set} is the current set value of the relay connected
R_R is the burden resistance of the relay

5.7.4.3 Instantaneous Function

$$V_{KP} \geq I_{sc} \times (R_{CT} + R_L + R_R),$$

Where in,
I_{sc} is the maximum short circuit current related to the secondary side of CT connected; also it is equal to the maximum fault current level divided by CT ratio.

5.7.4.4 Over-current Relay

$$V_{KP} \geq V_{KP\,(alreq)} = 2 \times I_{op} \times \frac{I_s}{I_p} \times (R_{CT} + R_L + Z_R)$$

Where in,
I_{op} is the primary operate value for Instantaneous and Definite Time value,
I_{op} is the maximum primary fault current for Inverse Time Delay
Is is the secondary rated current
Ip is the primary rated current
R_{CT} is the secondary resistance of CT
R_L is the resistance of wire
Z_R is the burden of the relay

5.7.5 Current Transformer Total Burden Requirement

The knee point voltage of CT connected with meters, i.e. ammeters, watt meters, volt meters, energy meters, etc. shall be based on the total burden of connected loads and calculated per formula below:

$P_B = P_L + P_R + P_{RCT}$, where in the following are defined as;

P_B is total CT connected burden, in VA
P_L is the total burden imposed by connecting lead wires, VA
P_R is the total burden imposed by meters and relays (if any)
P_{Rct} is the total internal burden imposed by CT secondary winding resistance (R_{CT})

The CT connected burden ($P_{B)}$ as per IEC-60044-1 shall be 25% -100% of the rated burden to maintain accuracy within specified limits.

6.0 INSTRUMENT POTENTIAL TRANSFORMERS

6.1 General

Instrument Potential Transformer (PT) steps down voltage of a circuit to a low value that can be effectively and safely used for operation of instruments such as, voltmeters, watt meters, and relays used for various protective purposes.
It is intended to have its primary winding connected in parallel with the power supply circuit, the voltage of which is to be measured or controlled. It should reproduce in its secondary circuit a definite ratio of voltage of its primary circuit with the phase relationship substantially preserved.

PTs are capable of continuous and accurate operation when the voltage applied across the primary at rated voltage is as follows:

a. 80% to 120% of rated voltage for measuring core PT at 80% power factor.
b. 5% of rated voltage for protection core PT and rated voltage multiplied by the voltage factor (1.2,1.5 or 1.9) at 80% power factor.

The standard values of rated primary voltage of a single-phase transformer connected between one line of a three-phase system and earth or between a system neutral point shall be 1/√3 times one of the values of rated system primary voltage.

The PT shall be generally inductive type with rated secondary voltage 110/√3 VAC or 120/√3 VAC or as specified in the data schedule for outdoor/indoor application.

6.2 PT Sizing for Protection and Metering

6.2.1 The standard accuracy class for protective potential transformers shall be 3P and 6P.

6.2.2 Accuracy class for measuring potential transformers shall be 0.3 as per IEEE C57.13 (2008) and 0.5 as per IEC 60044-2 (2003).

6.2.3 Applicable formulae for calculating the PT burden

The formula for calculating the PT total burden based on the connected meters and relays is as follow:

$$P_{Bvt} = P_{M1} + P_{M2} + P_{Mn} \ldots + P_{R1} + P_{R2} + P_{Rn}$$

Wherein,
P_{Bvt} is the total burden connected in voltage transformer, volt-ampere (VA)
P_{M1}, P_{M2}, P_{Mn} are the burden of each individual meters connected, VA
P_{R1}, P_{R2}, P_{Rn} are the burden of each individual relay connected, VA

Hence, Potential Transformer rating shall be approximately equal to

$PT_B \approx P_{M1} + P_{M2} + P_{Mn} \ldots + P_{R1} + P_{R2} + P_{Rn} + P_{OL}$

Wherein,

P_{OL}, other connected load burden, VA

PT_B, Voltage Transformer rating, VA

The Potential Transformer ratio shall be computed from the formula;

$PT_{Ratio} = Voltage_{primary} / Voltage_{secondary}$

Based on the equation for three-phase PT or single-phase PT used in single-phase system or between lines of a three-phase system, we get

$PT_{Ratio} = [System\ Nominal\ Voltage_{line\text{-}to\text{-}line\ primary}] / 120\ V$,

or

$PT_{Ratio} = [System\ Nominal\ Voltage_{line\text{-}to\text{-}line\ primary}] / 115\ V$

or for a single-phase PT connected between one line and earth (phase-to-earth) of a three-phase system, or between system neutral point, multiply both sides of the above formula by $1/\sqrt{3}$, Thus

$PT_{Ratio} = [System\ Nominal\ Voltage_{line\text{-}to\text{-}line\ primary} / \sqrt{3}] / 120/\sqrt{3}$

or

$PT_{Ratio} = [System\ Nominal\ Voltage_{line\text{-}to\text{-}line\ primary} / \sqrt{3}] / 115/\sqrt{3}$

Transformation ratio of VT primary and secondary voltage must be retained and whenever possible should be as simple value as indicated in IEC 60044-2.

6.2.4 Dual Class Type PT

If the burden consists of measuring and protection components, the higher accuracy class rating for the measuring equipment must be selected. For instance, to maintain accuracy of 0.5% for instruments when applied voltage is operating within 80% to 120% of rated voltage at rated frequency and maintains 3% accuracy for protection relays when voltage dipped down below 80% and up to 5% of the rated voltage, a dual class VT or CVT of 0.5/3P class is preferred.

The PT connected burden (PT_B) per IEC-60044-2, clause 12.2 and 13.2 shall be within 25% to 100% of the selected rating to maintain accuracy within specified limits.

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8.0 ANNEX 1 - SAMPLE OF CT SIZING CALCULATIONS

Part 1. Purpose and Design Basis

1. Scope

This section covers sample calculation of Current Transformer (CT) sizing for the metering and protection requirements of a substation and transmission line systems. The objective of this sample calculation is to present a uniform guidelines of sizing of current transformer to be adopted for, but not limited to the following:

a) 380kV GIS CTs for Underground Feeders
b) 380kV GIS CTs for Reactors
c) 380kV GIS CTs for high/low voltage side of (500MVA) Power Transformers
d) 380kV GIS CTs for Bus Coupler and
e) 380kV GIS CTs for Bus Section
f) 380kV GIS CTs for GIS Bus Bar
g) 380/132kV, 500MVA Power Transformer Bushing CTs
h) 380kV Reactor Bushings CTs

The values (e.g. CT burden, relay operating currents, etc.) are taken from manufacturer's catalogue for the purposes of arriving at the most realistic values during the computation.

2. Design basis

The following are the basis for CT sizing:

2.1 Current Transformer

- Breaker interrupting capacity as fault current
- R_{CT}, CT secondary winding resistance at 75ºC
- R_L, CT lead resistance at 75ºC
- 20% margin on calculated knee-point voltage
- 25% margin on calculated burden

2.2 System Data

- Nominal Voltage equal to 380 kV (presumed system voltage)
- System Short Circuit Current equal to 50 kA or 63 kA
- X/R Ratio equal to 50
- Primary DC time constant as 132.7 ms

2.3 CT Lead Data for stranded Copper Conductor Plain Wires as per IEC 60228

- DC resistance/m of 4mm^2 at 20ºC equal to 0.00461Ω/m
- DC resistance/m of 6mm^2 at 20ºC equal to 0.00308Ω/m
- DC resistance/m of 10mm^2 at 20ºC equal to 0.00183Ω/m

2.4 Max CT/VT Lead length … as per measurement

2.5 CT Calculation Criteria… as per relay manufacturer

Part II. CALCULATIONS for the following:

a. 380kV GIS CTs for 380kV Underground Feeders

The elements regarded for CT sizing are as below:

a.1. Relay to be connected

Single Phase High Impedance Differential Relays (one relay for each phase) shall be connected with the above CT.

Type of Relay: High Impedance Differential Relay (87-1) ABB Type RADHD

2. CT Data

CT Ratio	3000-2000/1
CT Class & Data	X, $V_{KP} \geq 450V$, $Im \leq 27mA$ @450V, $R_{CT} \leq 6\Omega$ $V_{KP} \geq 300V$, $Im \leq 40mA$ @300V, $R_{CT} \leq 4\Omega$

3. CT Lead Data

CT Lead Size	$10mm^2$
CT Lead Length	75m (one way distance considered for CTs)
CT Resistance/m	0.00183Ω/m at 20ºC

CT Lead Resistance at 75ºC for the 2 x 75 m = 150 m (for the two way lead wires)

Using the formula for resistance of CT leads,

$RL = R_{20} x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$, (CT Lead resistance at 75ºC)

Wherein,

$R_{20} \approx 0.00183\ \Omega/m$	(Resistivity of copper conductor of CT lead)
$L \approx 150$ m	(CT lead length)
$ά \approx 1/234.5$	(Coefficient of thermal expansion of copper conductor)

Substituting values;

$R_L = R_{20} x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$

$R_L = 0.339\ \Omega$

Relay to be connected: High Impedance Differential Relay (87-1) ABB Type RADHD

a.1.1. As per relay manufacturer recommendation the knee-point voltage (V_{KP}) requirement for the given type of relay shall be based on the condition that;

The knee-point voltage (V_{KP}) must be twice the voltage relay settings (Us), mathematically expressed as;

VKP $\geq$ 2 x IK2 x (RCT + RL), where

$I_{K2} = I_K/n$, (Maximum through fault current related to the secondary side of CT)

$I_K = 50$ kA (Maximum through fault current related to the primary side of CT)

n = 2000/1 (CT ratio)

$I_{K2} = I_K/n$

$I_{K2} = 25$ A

$R_{CT} = 4\ \Omega$ (CT secondary winding resistance)

Substituting values,

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 216.944$ V,

Applying 20% margin, (Based on SEC design based requirements)

$V_{KP1} = 1.2 \times V_{KP}$

VKP1 = 260.33V

Us $\approx V_{KP1}/2$ (Calculated Relay Setting Voltage)

Us $= V_{KP1}/2$
Us = 130.166 V

Us $\approx$ 200V (Proposed Relay Setting Voltage)

a.1.2. Effective Primary Operating Current

IP $\approx$ n x (ir + ires + $\sum$im + isup) (Relay primary operating current)

Where,

$i_r = 70$ mA	(Approximate RADHD relay type operating current at setting voltage, U_S) (Refer to ABB type RADHD relay publication)
$i_{res} = 3$ mA	(Current through the non-linear resistor at setting voltage "U_s")
$i_{sup} = 1.5$ mA	(RAEDK relay current at max. setting)
$i_m = 12$ mA	(CT magnetizing current at setting voltage "Us")
$\sum = 3$	(Total nos. of CT connected to Diffirential Protection)
n = 2000/1	(CT Ratio)

Substituting values to equation,

$I_P = n \times (i_r + i_{res} + \sum_{im} + i_{sup})$

$I_P = 221$ A

It is recommended that the sensitivity of the differential relay shall be 10- 20 % of the CT rated current at setting voltage.

$\%I = I_P / I_{CT} \times 100$

$\%I = 11.05$

The primary operating current is within recommended value. However, as per SEC requirement the primary operating current shall be 120% of the full load current at setting voltage.

To increase the primary operating current to 120% of the full load current, a variable shunt resistor shall be used.

a.1.3. Calculating for the required shunt resistor;

$I_P \approx n \times (i_r + I_{res} + \sum \times i_m + i_{sup})$

$I_P = 221$ A (Total primary pickup current)

The primary pickup current should be such that with opening of one of the CT connected with connection protection and loading of that feeder equal to the CT rated current i.e. 2000 A. (approx. 120% F.L. current) should not cause tripping of the connection protection.

$I_{P1} = 2000$A

The primary current 2000 A has been adopted as primary operate current for both differential protections.

As the primary pick up current is less than the proposed limit. It can be increased by addition of shunt resistor R_{sh}

$I_{op} = n \times (i_r + i_{res} + \sum \times i_m + i_{sup}) + n \times I_{sh}$

$$I_{sh} = \frac{I_{P1} - N \times (i_r + i_{res} + \sum \times i_m + i_{sup})}{n}$$

$I_{sh} = 0.889$ A

$R_{sh} = U_S/I_{sh}$ (Resistance of the shunt resistor)
$R_{sh} = 224.845\Omega$

The Relay Setting

Setting Type	Range	Calculated Settings	Adopted Settings
Operate Voltage	100-400V	130V	200V

The available/computed knee point voltage is more than the required knee point voltage. Hence the CT Core selected is suitable for the required protection application.

b. 380kV GIS CTs for 380kV Reactors

b.1 Proposed relay to be connected

Single Phase High Impedance Differential Relays (one relay for each phase) shall be connected with the below mentioned CT.

Type of Relay assumed: High Impedance Differential Relay (87R-1) ABB Type RADHD (this relay was chosen for computing purposes only)

Proposed CT Data

CT Ratio	600-300/1
CT Class & Data	X, $V_{KP} \geq 600$V, Im $\leq$ 150mA @600V, $R_{CT} \leq 1.6\Omega$ $V_{KP} \geq 300$V, Im $\leq$ 300mA @300V, $R_{CT} \leq 0.8\Omega$

CT Lead Data

CT Lead Size	$4mm^2$

CT Lead Length	110m (one way distance considered for BCTs of K505 Shunt Reactor as the worst case)
CT Resistance/m	0.00461Ω/m at 20ºC

CT Lead Resistance at 75ºC for the 2 x 110m = 220 m (for the two way lead wires)

Using the formula for resistance,

$RL = R_{20} x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$, (CT Lead resistance at 75ºC)

wherein,

$R_{20} \approx 0.00461\ \Omega/m$	(Resistivity of copper conductor of CT lead)
$L \approx 220$ m	(CT lead length)
$ά \approx 1/234.5$	(Coefficient of thermal expansion of copper conductor)

Substituting values;

$R_L = R_{20}\ x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$

$R_L = 1.252\ \Omega$

<u>High Impedance Differential Relay (87-1) ABB Type RADHD</u>

b.1.1. As per relay manufacturer recommendation the knee point voltage (V_{KP}) requirement for the given type of relay shall be following:

VKP $\geq$ 2 x IK2 x (RCT + RL),
where

$IK \approx 10\ x\ \frac{SrS}{\sqrt{3}\ x\ Ur,}$	(Maximum through fault current related to primary side of CT has considered as inrush current of Shunt Reactor (SR) which shall be 10 times the rated current of SR)
$S_{rS} = 80\ x\ 10^6$ V x A	(SR rating in VAR has been considered as VA)

$I_K \approx 10\ x\ \frac{Srs}{\sqrt{3}\ x\ U_r}$

$I_K = 1.215\ x\ 10^3$ A	(Maximum through fault current related to the primary side of CT)
n = 600/1	(CT ratio)

$I_{K2} = I_K/n$ (Maximum through fault current related to the secondary side of CT)

$I_{K2} = 2.026$ A (Maximum through fault current related to the primary side of CT)

$R_{CT} = 1.6\ \Omega$ (CT secondary winding resistance)

Substituting values,

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 11.555$ V,

Applying 20% margin will give us,

$V_{KP1} = 1.2 \times V_{KP}$

$V_{KP1} = 13.866$ V

$Us \approx V_{KP1}/2$ (Proposed relay setting voltage)

$Us = V_{KP1}/2$

$Us = 6.933$ V

$Us \approx 50$V (Proposed relay setting voltage)

b.1.2. Primary Operate Current

$IP \approx N \times (ir + Ires + \sum im + isup)$ (Relay primary operate current)

Where,

$Ir = 70$mA (Approximate relay type operating current at setting voltage U_S)

$I_{res} = 0$mA (Current through the non-linear resistor at setting voltage "U_s")

$I_{sup} = 1.5$mA (RAEDK relay current at max. setting)

$Im = 12$mA (CT magnetizing current at setting voltage "Us")

$\sum = 2$ (Total nos. of CT connected to Diff. Prot.)

$N = 600/1$ (CT ratio)

$I_P = N \times (i_r + I_{res} + \sum_{im} + i_{sup})$

$I_P = 57.3$ A

It is recommended that the sensitivity of the differential relay shall be 10- 20 % of the CT rated current at setting voltage.

$I_{CT} = 600$ A

$\%I = I_P / I_{CT} \times 100$

$\%I = 9.55$

Based on the result, the primary operate current is near to the recommended value.

The available knee point voltage is more than the required knee point voltage. Hence, the selected CT core is suitable for the required protection application.

b.2. Protection Relay + Meters to be connected

• Digital Multimeter Power Measurement type	–	0.01 VA
• Current Transducer	-	0.20 VA
• Watt/Var transducer	-	0.20 VA
• 50 BF (1) Relay	-	0.02 VA
• 50 BF (2) Relay	-	2.64 VA
• 51/51N (O/C) Relay	-	0.10 VA
Total Burden	=	3.17 VA

Overload rating of Power Measuring Instruments = 120 A for 1 sec
Overload rating of Current Transducer = 10 A for 1 sec

<u>Proposed CT Data</u>

CT Ratio	600-300/1
CT Class & Data	5P20/1.0, 60-30 VA
CT secondary resistance	$R_{CT} = 1.6 - 0.8\Omega$

<u>CT Lead Data</u>

CT Lead Size	$4mm^2$
CT Lead Length	155m (one way distance considered for BCTs of K04 Shunt Reactor as the worst case)
CT Resistance/m	0.00461Ω/m at 20ºC

CT Lead Resistance at 75ºC for the 2 x 155m = 310 m (two way)

Using the formula for resistance,

$R_L = R_{20} \times L \times [\ 1 + \acute{\alpha} \times (\ 75 - 20\ )\ ]$, (CT Lead resistance at 75ºC)

wherein,

$R_{20} \approx 0.00461\ \Omega/m$ (Resistivity of copper conductor of CT lead)

$L \approx 310$ m (CT lead length)

$\acute{\alpha} \approx 1/234.5$ (Coefficient of thermal expansion of copper conductor)

Substituting values;

$R_L = R_{20} \times L \times [\ 1 + \acute{\alpha} \times (\ 75 - 20\ )\ ]$

$R_L = 1.764\ \Omega$

b.2.1. *To verify that accuracy of the CT shall be maintained:*

As per IEC 60044-1 (Clause 11.2) for CTs with accuracy 1.0 as in the subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy within specified limits.

$I_n = 1$ A

$P_{LO} = I_n^2 \times R_{LO}$ (Lead wires burden)
$P_{LO} = 1.764$ W

$P_M = 3.17$ V x A (Instrument, i.e. (relay + meter) burden)

$P_B = P_M + P_{LO}$ (Total burden on CT)

$P_B = 4.934$ W

PN = 60 V x A (Rated burden of CT)

$\%P = \frac{P_B}{P_N} \times 100$ (% of connected burden of CT)

$\%P = 8.057$

As the connected burden is less than 25% of the rated burden, the accuracy of the CT cannot be maintained within specified limits. To get the required accuracy an Interposing Current Transformer (IPCT / auxiliary CT) has to be installed.

IPCT Data

CT Ratio	1/1
CT Class	$1.0F_S5$, 3VA
CT secondary resistance	1.32 Ω
I_{th} for 1 sec	100A

CT Lead Data

Distance from IPCT to meters in control panel

CT Lead size	2.5 mm^2
CT Lead length	5m (one way distance while considering IPCT in Control Panel
CT Resistance	0.00741 Ω/m at 20ºC

Computing for CT Lead resistance at 75ºC @ 10 m (two way length)

$RL = R_{20} \times L \times [1 + ά \times (75 – 20)]$ (CT Lead resistance at 75ºC)

$R_{20} \approx 0.00741$ Ω/m (Resistivity of copper conductor of CT lead)

$L_1 \approx 10$ m (CT lead length)

$ά \approx 1/234.5$ (Coefficient of thermal expansion of copper conductor)

Substituting values;

$R_{L1} = R_{20} \times L \times [1 + ά \times (75 – 20)]$

$R_{L1} = 0.091$ Ω

$In_1 = 1$ A (IPCT secondary current)

$P_{L1} = I_n^2 \times R_{L1}$ (Lead burden from IPCT to meters in CP)

$P_{L1} = 0.091$ V x A

CT Lead Data

From installed meters in CP Transducer Panel

CT Lead Size $4mm^2$

CT Lead length	40m (one way distance while considering IPCT in Control Panel
CT Resistance	0.00461 Ω/m at 20ºC

CT Lead resistance at 75ºC @ 80 m (two way length)

$RL = R_{20} x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$, (CT Lead resistance at 75ºC)

$R_{20} \approx 0.00741\ \Omega/m$ (Resistivity of copper conductor of CT lead)

$L_3 \approx 80$ m (CT lead length)

$ά \approx 1/234.5$ (Coefficient of thermal expansion of copper conductor)

Substituting values;

$R_{L3} = R_{20} x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$

$R_{L3} = 0.455\ \Omega$

$In_1 = 1$ A (IPCT secondary current)

$P_{L3} = I_n^2\ x\ R_{L3}$ (Lead burden from meters in CP to transducers)

$P_{L1} = 0.455W$

$P_M = 0.41$ V x A (Meter burden)

$P_{BC1} = P_M + P_{L1} + P_{L3}$ (Total burden on IPCT)
$P_{BC1} = 0.956$ W

PBN1 = 3V x A (Rated burden of IPCT)

As per IEC 60044-1 (Clause 11.2) for CTs with accuracy 1.0 as in the subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy within specified limits.

$$\%P = \frac{P_{BC1}}{P_{BN1}} x\ 100$$ (% of connected burden of CT)

%P = 31.877

As the connected burden is more than 25% of the rated burden of IPCT, the accuracy of IPCT can be maintained within specified limits.

A. To verify that the instruments shall be safe under fault condition:

Operating instrument security safety factor (K_{OFS}) < Rated Instruments security factor (K_{RFS}) of instruments

$R_{CT1} = 1.32\ \Omega$ (IPCT secondary resistance)

$P_{I1} = I_{n1}^2 \times R_{CT1}$ (Internal burden of IPCT)

$P_{I1} = 1.32$ W

$P_{BC1} = 0.956$ W (Total burden of CT)

With a 25% margin

$P_{BC1'} = 1.25 \times P_{BC1}$

PBC1' = 1.195 W

$K_{nFs1} = 5$ (Nominal IPCT instrument security factor)

$K_{oFs1} = K_{nFs1} \times \frac{[PBN1 + PI1]}{[PI1 + P_{BC1'}]}$ (Operating accuracy limiting factor)

$K_{oFs1} = 8.587$

$K_{RFs1} = \frac{I_{SC1}}{I_{n1}}$ (Rated Instrument security factor of instruments)

$Isc_1 = 40$ A (Short time current rating of instruments, value is taken from manufacturer of instrument)

$I_{n1} = 1$ A (Nominal current rating of instruments)

$K_{RFs1} = 40$

Based on the condition that $K_{oFs1} < K_{RFs1}$, The available operating instrument security factor is less than rated instrument security factor of instrument. Hence, the proposed IPCT data is suitable for required metering application.

B. To verify that the instruments shall be safe under fault condition:

- The thermal rating of the instruments shall be comparable with the thermal rating of the IPCT.
- The Instruments, i.e. Power Measurements, Current and Transducers are having the following short time rating:

$120 \times I_n / 1s$, $10 \times I_n / 10_s$, $40 \times I_n / 1s$

The thermal rating of the weakest instrument connected shall:

$I_m \approx 10 \times I_{n1}$ (Overload rating of the meter (A))

$t_m = 10$ sec. (Time period for which the meters can be overloaded, second)

$K_m = I_m^2 \times t_m$ (Thermal rating of the meters)

Thermal rating of IPCT at operating instrument security factor:

$I_{IPCT} = K_{OFSL} \times I_{n1}$ (IPCT current in ampere at operating instrument security factor)

$t_{IPCT} = 1$ sec. (IPCT Thermal withstand time, second)

$K_{IPCT} = I_{IPCT}^2 \times t_{IPCT}$ (Thermal rating of IPCT at operating instrument security factor)

At IPCT operating instrument security factor, the meter should withstand the fault current for the time specified below:

$$t_{mR} = \frac{I_{IPCT}^2 \times t_{IPCT}}{I_m^2}$$ (Required withstand time of meters under fault cond.)

$$t_{mR} = \frac{[K_{OFSL} \times I_{n1}]^2 \times t_{IPCT}}{I_m^2}$$

$t_{mR} = 0.737$ sec (Required withstand time of meters)

The Factor of Safety (FOS) may be calculated as follows:

$K_{mR} = I_m^2 \times t_{mR}$ (Required thermal rating of meter)

$$FOS = \frac{K_m}{K_{mR}}$$

FOS = 13.561

The required time for instruments to withstand against the fault is less than the withstand capability of the instruments. Hence, the instruments are suitable for required metering application.

C. To verify that the IPCT shall be safe under fault condition against main CT :

For the safety of IPCT during fault condition, the thermal rating of the IPCT shall be comparable with the thermal rating of the main CT.

The IPCT proposed short time rating:

100A for 1 sec.

The thermal rating of the IPCT:

$I_{PCT} = 100$ A (Thermal rating of IPCT)

$t_{IPCT} = 1$ sec (Time or period for which the IPCT can be overloaded without suffering harmful effect)

$K_{IPCT} = I_{IPCT}^2 \times t_{IPCT}$ (Thermal rating of IPCT)

$K_{IPCT} = 1.0 \times 10^4 \ A^2$

The main CT is having the following short time rating:

100kA for 1 sec. as per SEC data schedule of GIS CT. However, the max. fault current shall be 50kA for 1 sec. Hence, value of I_{th} for CT shall be taken according to max. fault current.

The thermal rating of the CT shall be as follows:

$I_{th} = 50 \times 10^3$ (Primary thermal rating of the CT)

$I_N = 600$ (Primary rated current of the CT)

$$I_{ths} = \frac{I_{th}}{I_N}$$

$t_{CT} = 1$ (Time period, in sec., for which the CT can be overloaded without suffering from harmful effect)

$K_{CT} = I_{ths}^2 \times t_{CT}$ (Thermal rating of the CT)

At main CT thermal overloading, the IPCT should withstand the fault current for the time specified below:

$$t_{IPCTR} = \frac{I_{ths}^2 \times t_{CT}}{I_{IPCT}^2}$$ (Required withstand time of meters under fault cond.)

$t_{IPCT} = 0.694$ sec.

The FOS may be calculated as follows:

$K_{IPCTR} = I_{IPCT}^2 X\ t_{IPCTR}$ (Required thermal rating of meter)

$$FOS = \frac{K_{IPCT}}{K_{IPCTR}}$$

FOS = 1.44

The required time for IPCT to withstand against the fault is less than the withstand capability of the IPCT. Hence, the IPCT are suitable for required metering application.

D. Calculation for Protection Relays:

To ensure that CT will not be saturated under fault conditions:

Operating accuracy limiting factor (K_{OALF}) > Required accuracy limiting factor (K_{RALF})

CT Lead Data
From main CT to control panel

CT Lead Size	$4mm^2$
CT Lead length	110m (one way lead dist. considering IPCT in control panel)
CT resistance	0.00461 Ω/m at 20ºC

CT Lead resistance at 75 ºC @ 220m (two way)

$RL2 = R_{20} x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$, (CT Lead resistance at 75ºC)

$R_{20} \approx 0.00461\ \Omega/m$ (Resistivity of copper conductor of CT lead)

$L_2 \approx 220$ m (CT lead length)
$ά \approx 1/234.5$ (Coefficient of thermal expansion of copper conductor)

Substituting values;

$R_{L2} = R_{20} x\ L\ x\ [\ 1 + ά\ x\ (\ 75 – 20\ )\ ]$

$R_{L2} = 1.252\ \Omega$

$I_{n2} = 1$ A (Main CT secondary current)

$P_{L2} = I_{n2}^2\ x\ R_{L2}$ (Lead burden from the meters to transducers)

PL2 = 1.252 W

RCT11 = 1.32 Ω [IPCT primary side resistance (based from manufacturer)]

Ptp = In22 x RCT11

Ptp = 1.32 W

PR = 2.76 V x A, or 2.76 W (Total relay burden)

PBC1 = 0.956 W (Total Burden of CT)

Substituting values to equation, will result to

$P_B = P_R + P_{L2} + P_{BC1r} + P_{tp}$ (Total burden of CT)

$P_B = 6.288$ W (Total burden of CT)

With 25% margin,

$P_{B1} = 1.25 \times P_B$

PB1 = 7.86 W

$R_{CT} = 1.6\ \Omega$

Internal Burden of CT

$P_I = I_n^2 \times R_{CT}$

$P_I = 1.6$ W

To ensure CT does not saturate:

Operating accuracy limit factor (K_{OALF}) > Required accuracy limit factor (K_{ALFR})

$$K_{OALF} = K_{ALF} \times \frac{[P_N + P_I]}{[P_1 + P_B]}$$ (Operating limiting factor)

$K_{ALF} = 20$ (CT accuracy limiting factor)

$P_N = 60V \times A$ (Rated CT Burden)

$$K_{OALF} = K_{ALF} \times \frac{[P_N + P_I]}{[P_1 + P_B]}$$ (Operating limiting factor)

$K_{OALF} = 156.19$

$$K_{ALFR} = \frac{I_{sc}}{I_P}$$

I_{sc} = 50 x 10^3 A

I_P = 600 A

K_{ALFR} = 83.333

As, K_{OALF} > K_{ALFR}, then

This satisfy the condition that the available operating accuracy limiting factor is more than required accuracy limit factor of CT. Hence the proposed current transformer (CT4-3) data is suitable for required protection application.

c. 380kV GIS CTs for 380kV side of 500 MVA Transformers

Relay to be connected:

Following single phase high impedance differential relay (one relay for each phase) shall be connected with the above mentioned CT.

High impedance transformer differential relay (87T-1) ABB Type RADHD.

PROPOSED CT Data:

CT Ratio	3000/1
CT Class & data	$X, V_{KP} \geq 400V, I_m \leq 30mA@400V,$ $R_{CT} \leq 5\Omega$

CT Lead Data:

CT Lead size	4mm^2
CT Lead length	115m (one way distance for SGT3 as worst case)
CT Resistance/m	0.00461Ω/m at 20°C

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 230m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$

$R_L = 1.309\Omega$

c.1. As per relay manufacturer recommendation the knee point voltage (V_{KP}) requirement for RADHD relay is the following:

$V_{KP} \geq 2 \times I_{K2} \times (R_{CT} + R_L)$

Where

$I_{K2} = \frac{I_K}{n}$ (Maximum through fault current related to the secondary side of CT)

$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n \times u_{kr}}$ (Maximum through fault current related to the primary side of CT)

$S_{rT} = 500 \times 10^6 V \times A$ (Rated transformer power)

$U_n = 380kV$

$u_{kr} = 19\%$ (Rated transformer impedance voltage at principal tap)

$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n \times u_{kr}}$

$I_K = 3.998 \times 10^3 A$

$n = \frac{3000}{1}$ (CT Ratio)

$I_{K2} = \frac{I_K}{n}$

$I_{K2} = 1.333A$

$R_{CT} = 5\Omega$ (CT secondary winding resistance)

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 16.817V$

(With 20% margin)

$V_{KP1} = 120\% \times V_{KP}$

$V_{KP1} = 20.18V$

c.2. Considering busbar fault current as the maximum through fault currents the knee point voltage (V_{KP}) requirement for RADHD relay is the following:

$V_{KP} \geq 2 \times I_{K2} \times (R_{CT} + R_L)$

Where

$I_{K2} = \frac{I_K}{n}$ (Maximum through fault current related to the secondary side of CT)

$I_K = 50kA$ (Maximum through fault current related to the primary side of CT)

$n = \frac{3000}{1}$ (CT Ratio)

$I_{K2} = \frac{I_K}{n}$

$I_{K2} = 16.667A$

$R_{CT} = 5\Omega$ (CT secondary winding resistance)

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 210.299V$

(With 20% margin)

$V_{KP11} = 1.2 \times V_{KP}$

$V_{KP11} = 252.359V$

c.3. As per SEC recommendation, 16 times the rated transformer current consider as the maximum through fault currents, the knee point voltage (V_{KP}) requirement for RADHD relay is the following:

$V_{KP} \geq 2 \times I_{K2} \times (R_{CT} + R_L)$

Where

$I_{K2} = \frac{16I_K}{n}$ (Maximum through fault current related to the secondary side of CT)

$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n}$ (Maximum through fault current related to the primary side of CT)

$S_{rT} = 500 \times 10^6 V \times A$ (Rated transformer power)

$U_n = 380kV$

$$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n}$$

$I_K = 759.671A$

$$n = \frac{3000}{1}$$ (CT Ratio)

$$I_{K2} = \frac{16 I_K}{n}$$

$I_{K2} = 4.052A$

$R_{CT} = 5\Omega$ (CT secondary winding resistance)

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 51.123V$

(With 20% margin)

$V_{KP1} = 1.2 \times V_{KP}$

$V_{KP1} = 61.347V$

$$U_S = \frac{V_{KP11}}{2}$$ (Setting voltage)

$U_S = 126.18V$

$U_S = 150V$ (Proposed setting voltage)

c.4. Primary operate current:

$I_P = N \times (i_r + i_{res} + \sum i_m + i_{\sup})$ (Relay primary operate current)

Where

$i_r = 70mA$ (Approximate RADHD relay operating current at setting voltage 'U$_S$')

$i_{res} = 1mA$ (Current through the non-linear resistor at setting voltage 'U$_S$')

$i_{sup} = 1.5mA$ (RAEDK relay current at max. setting)

$i_m = 12mA$ (CT megnatizing current at setting voltage 'U$_S$')

$\sum = 3$ (Total number of CTs connected to Differential Protection)

$N = \frac{3000}{1}$ (CT Ratio)

$$I_P = N \times (i_r + i_{res} + \sum \times i_m + i_{sup})$$

$$I_P = 325.5A$$

It is recommended that the sensitivity of the differential relay shall be 10-20% of the CT rated current at setting voltage.

$$I_{CT} = 3000A$$

$$\%I = \frac{I_P}{I_{CT}} \times 100$$

$$\%I = 10.85$$

The primary operate current is within recommended value.

The available knee point voltage is more than required knee point voltage. Hence the selected CT core is suitable for required protection application.
Relay to be connected and their data:

- O/C (50P/51P) Relay ABB Type REJ525 :0.1VA
- 50BF (1) Relay ABB Type RAICA :0.02VA
- 50BF (2) Relay Areva Type MCT140 :2.64VA

Total Burden :2.76VA

CT Data

CT Ratio	3000/1
CT Class & Data	5P20, 15VA
CT Secondary Winding Resistance (R_{CT})	5Ω

CT Lead Data

CT Lead size	$4mm^2$
CT Lead length	65m (one way distance for K02 as worst case)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2× 65 = 130m (two way)

$R_L = R_{20} \times L \times \left[1 + \alpha \times (75 - 20)\right]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 130m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times \left[1 + \alpha \times (75 - 20)\right]$

$R_L = 0.74\Omega$

$I_n = 1A$

$P_L = I_n^{\,2} \times R_L$

$P_L = 0.74W$

$R_{CT} = 5\Omega$

$P_I = I_n^{\,2} \times R_{CT}$ (Internal burden of CT)

$P_I = 5W$

$P_B = P_R + P_L$ (Total burden on CT)

$P_R = 2.76V \times A$ (Relay burden)

$P_B = P_R + P_L$

$P_B = 3.5W$

With 25% margin

$P_{B1} = 125\% \times P_R$

$P_{B1} = 4.375W$

To ensure CT does not saturate:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{ALFR})

$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$ (Operating accuracy limit factor)

$K_{ALF} = 20$ (CT accuracy limit factor)

$P_N = 15V \times A$ (Rated CT burden)

$$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$$

$K_{OALF} = 42.667$

$$K_{ALFR} = \frac{I_{SC}}{I_P}$$

$I_{SC} = 50 \times 10^3 A$

$I_P = 3000A$

$$K_{ALFR} = \frac{I_{SC}}{I_P}$$

$K_{ALFR} = 16.667$

As $K_{OALF} > K_{ALFR}$
The available operating accuracy limit factor is more than required accuracy limit factor. Hence the proposed CT6-1 data is suitable for required protection application.

Device to be connected and its data:

- Digital multimeter power measurement type 6200 :0.01VA
- Energy meter alpha (AIR) :0.0001VA
- Current Transducer CEWE Type DI02 :0.2VA
- Watt/Var Transducer CEWE Type DQ34 :0.2VA

Total Burden :0.4101VA

Overload rating Power measurement instruments :120A for 1 sec
Overload rating CEWE instruments :10A for 10 sec

CT Data

CT Ratio	3000-2000-1000/1
CT Class	1.0Fs5, 30-15-10VA
CT Secondary Resistance	7-3.5-2.5Ω

CT Lead Data

CT Lead size	$4mm^2$
CT Lead length	80m (one way distance for K03 as worst case)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2× 80 =160m (two way)

$R_{L0} = R_{20} \times L \times \left[1 + \alpha \times (75 - 20)\right]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_0 = 160m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L0} = R_{20} \times L_0 \times \left[1 + \alpha \times (75 - 20)\right]$

$R_{L0} = 0.911\Omega$

c.4.1 To verify that accuracy of CT shall be maintained:

As per IEC 60044-1 for CTs with accuracy 1.0%, as in subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy with the specified limits.

$I_n = 1A$

$P_{L0} = I_n^{\ 2} \times R_{L0}$ (Lead burden)

$P_{L0} = 0.911W$

$R_M = 0.4101 \times A$ (Meter burden)

$P_B = P_M + P_{L0}$ (Total burden on CT)

$P_B = 1.321W$

$R_N = 10V \times A$ (Rated burden of CT)

$\%P = \frac{P_B}{P_N} \times 100$ (% of connected burden of CT)

$\%P = 13.207$

As the connected burden is less than 25% of the rated burden, the accuracy of the CT cannot be maintained within specified limits. To get the required accuracy an IPCT has to be installed.

IPCT Data

CT Ratio	1/1
CT Class	1.0Fs5, 3VA
CT Secondary Resistance	1.32Ω
I_{th} for 1 sec.	50A

CT Lead Data
From IPCT to meter in Control Panel

CT Lead size	2.5mm^2
CT Lead length	5m (one way distance while considering IPCT in control panel)
CT Resistance/m	0.00741Ω/m at 20°C

CT lead resistance at 75°C =10m (two way length)

$R_{L1} = R_{20} \times L_1 \times \left[1 + \alpha \times (75 - 20)\right]$ (CT lead resistance at 75°C)

$R_{20} = 0.00741 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_1 = 10m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L1} = R_{20} \times L_1 \times \left[1 + \alpha \times (75 - 20)\right]$

$R_{L1} = 0.091\Omega$

$I_{n1} = 1A$ (IPCT secondary current)

$P_{L1} = I_{n1}^{2} \times R_{L1}$ (Lead burden from IPCT to meters in Control Panel)

$P_{L1} = 0.091W \times A$

CT Lead Data
From meters in CP to Transducer Panel

CT Lead size	$4mm^2$
CT Lead length	40m (one way distance while considering IPCT in control panel)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C 80m (two way length)

$R_{L3} = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_3 = 80m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L3} = R_{20} \times L_3 \times [1 + \alpha \times (75 - 20)]$

$R_{L3} = 0.455\Omega$

$I_{n1} = 1A$ (IPCT secondary current)

$P_{L3} = I_{n1}^{2} \times R_{L3}$ (Lead burden from IPCT to meters in Control Panel)

$P_{L3} = 0.455W$

$R_M = 0.4101 \times A$ (Meter burden)

$P_{BC1} = P_M + P_{L1} + P_{L3}$ (Total burden on CT)

$P_{BC1} = 0.956W$

$R_{BN1} = 3V \times A$ (Rated burden of CT)

As per IEC 60044-1 for CT with accuracy 1.0, as in subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy with the specified limits.

$\%P = \frac{P_{BC1}}{P_{BN1}} \times 100$ (% of connected burden of CT)

$\%P = 31.88$

As the connected burden is more than 25% of the rated burden of IPCT, the accuracy of IPCT cannot be maintained within specified limits.

c.4.2.1 To verify that instruments shall be safe under fault condition:

Operating instrument security factor (K_{0FS}) < Rated instrument security factor (K_{RFS} of instrument)

$R_{CT1} = 1.32\Omega$ (IPCT secondary burden)

$P_{I1} = I_{n1}^{2} \times R_{CT1}$ (Internal burden of IPCT)

$P_{I1} = 1.32W$

$P_{BC1} = 0.956W$ (Total burden on IPCT)

With 25% margin

$P_{BC1} = 125\% \times P_{BC1}$

$P_{BC1} = 1.195W$

$K_{nFs1} = 5$ (Nominal IPCT instrument security factor)

$K_{oFs1} = K_{nFs1} \times \frac{P_{BN1} + P_{I1}}{P_{I1} + P_{BC1}}$ (Operating accuracy limit factor)

$K_{oFs1} = 8.587$

$K_{RFs1} = \frac{I_{sc1}}{I_{n1}}$

$I_{sc1} = 40A$

$I_{n1} = 1A$

$K_{RFs1} = 40$

As $K_{oFs1} < K_{RFs1}$

The available operating instrument security factor is less than rated instrument security factor of instruments. Hence the proposed IPCT data is suitable for required metering application.

c.4.2.2 To verify that the instruments shall be safe under fault condition:

The thermal rating of the instruments shall be comparable with the thermal rating of the IPCT.

The instruments i.e. Power Measurement make and CEWE make are having the following short time rating:

120 x In /Is & 10 x In /10s ~ 40 x In /Is

The thermal rating of the weakest instrument connected:

Im = 10 x In1	(Overload rating of the meter in amperes)
tm = 10	(Time period in second for which the meters can be overloaded)
$Km = Im^2 \times tm$	(Thermal rating of the meters)

Thermal rating of the IPCT at operating instrument security factor:

$I_{IPCT} = K_{0Fs1} \times In1$

$t_{IPCT} = 1$	(IPCT thermal withstand time in second)
$K_{IPCT} = I_{IPCT}^2 \times t_{IPCT}$	(Thermal rating of IPCT at operating instrument limit factor)

At IPCT operating instrument security factor, the meter should withstand the fault current for the time specified below:

$t_{mR} = \frac{I_{IPCT}^2 \times t_{IPCT}}{I_m^2}$	(Required withstand time of meters under fault conditions)
$t_{mR} = 0.737$	(Required withstand time of meters in second)

The factor of safety 'FoS' may be calculated as follows:

$K_{mR} = I_m^2 \times t_{mR}$	(Required thermal rating of the meter)

$$FoS = \frac{K_m}{K_{mR}}$$

FoS = 13.563

The required time for instruments to withstand against the fault is less than the withstand capability of the instruments. Hence the instruments are suitable for required metering application.

c.4.3. To verify that IPCTs shall be safe under fault condition against main CT:

$P_{BC2} = P_{BC1r} + P_{L2} + P_{tP}$ (Reflected burden on main CT)

$P_{BC1r} = \frac{P_{BC1} + P_{ts}}{n^2}$ (Reflected burden of IPCT, its circuit and relays)

$n = \frac{1}{1}$ (IPCT ratio)

$P_{ts} = I_{n1}^{\ 2} \times R_{CT1}$

P_{ts} = 1.32 W

$P_{BC1r} = \frac{P_{BC1} + P_{ts}}{n^2}$ (Total burden on IPCT)

P_{BC1r} = 2.515 W

CT Lead Data
From main CT to control panel

CT Lead size	$4mm^2$
CT Lead length	35m (one way distance while considering IPCT in control panel)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C 80m (two way length)

$R_{L2} = R_{20} \times L_2 \times \left[1 + \alpha \times (75 - 20)\right]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_2 = 70m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L2} = R_{20} \times L_2 \times \left[1 + \alpha \times (75 - 20)\right]$

$R_{L2} = 0.398\Omega$

$I_{n2} = 1A$ (Main CT secondary current)

$P_{L2} = I_{n2}^{\ 2} \times R_{L2}$ (Lead burden from meters to transducers)

$P_{L2} = 0.398W$

$R_{CT11} = 1.32\Omega$ (IPCT primary side resistance)

$P_{tP} = I_{n2}^{\ 2} \times R_{CT11}$

$P_{tP} = 1.32W$

$P_{BC2} = P_{BC1r} + P_{L2} + P_{tP}$

$P_{BC2} = 4.234W$

$P_{BC2} = 125\% \times P_{BC2}$

$P_{BC2} = 5.292W$

$R_{CT2} = 2.5\Omega$ (Main CT secondary resistance)

$P_{I2} = I_{n2}^{\ 2} \times R_{CT2}$

$P_{I2} = 2.5W$

$P_{BN2} = 10V \times A$ (Main CT rated burden)

$K_{nFs2} = 5$ (Nominal main CT instrument security factor)

$$K_{oFs2} = K_{nFs2} \times \left(\frac{P_{BN2} + P_{I2}}{P_{I2} + P_{BC2}}\right)$$

$K_{oFs2} = 8.021$

For the safety of IPCT during fault condition, the thermal rating of the IPCT shall be comparable with the thermal rating of the main CT.

The IPCT proposed short time rating:

50A for 1 second

The thermal rating of the IPCT:

$I_{IPCT} = 50$ (Thermal rating of the IPCT in amperes)

$t_{IPCT} = 1$ (Time period in second for which the IPCT can be overloaded without suffering harmful effect)

$K_{IPCT} = I_{IPCT}^{2} \times t_{IPCT}$ (Thermal rating of the IPCT)

The main CT is having the following short time rating:

The thermal rating of the main CT:

$I_{n2} = 1$ (Main CT secondary current in amperes)

$I_{th} = K_{oFs2} \times I_{n2}$ (Primary thermal rating of the CT in amperes)

$t_{CT} = 1$ (Time period in second for which the CT can be overloaded without suffering harmful effect)

$K_{CT} = I_{th}^{2} \times t_{CT}$ (Thermal rating of the CT)

At main CT thermal overloading, the IPCT should withstand the fault current for the time specified below:

$t_{IPCTR} = \frac{I_{th}^{2} \times t_{CT}}{I_{IPCT}^{2}}$ (Required withstand time of meters under fault conditions)

$t_{IPCTR} = 0.026$ (Required withstand time of IPCT in seconds)

The factor of safety 'FoS' may be calculated as follows:

$K_{IPCTR} = I_{IPCT}^{2} \times t_{IPCTR}$ (Required thermal rating of the meter)

$$FoS = \frac{K_{IPCT}}{K_{IPCTR}}$$

$$FoS = 38.861$$

The required time for IPCT to withstand against the fault is less than the withstand capability of the IPCT. Hence the IPCTs are suitable for required metering application.

d. 380KV GIS CTs for 380kV Bus Coupler

Device to be connected and its data:

- Digital multimeter power measurement type 6200 :0.01VA
- 50BF (1) Relay ABB Type RAICA :0.02VA
- 50BF (2) Relay Areva Type MCT140 :2.64VA
- O/C (50P/51 & 50/51N) Relay ABB Type REJ525 :0.1VA

Total Burden :2.77VA

Overload rating for all metering instruments :120A x I_n

CT Data

CT Ratio	4000/1
CT Class	5P20/1.0, 15VA
CT Secondary Resistance	5Ω

CT Lead Data

CT Lead size	$4mm^2$
CT Lead length	60m (one way distance as a worst case for K130 has been taken)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2× 60 =120m (two way)

$$R_{L0} = R_{20} \times L_0 \times \left[1 + \alpha \times (75 - 20)\right]$$ (CT lead resistance at 75°C)

$$R_{20} = 0.00461 \frac{\Omega}{m}$$ (Resistivity of copper conductor of CT lead)

$$L_0 = 120m$$ (CT lead length)

$$\alpha = \frac{1}{234.5}$$ (Co-efficient of thermal expansion of copper conductor)

$$R_{L0} = R_{20} \times L_0 \times \left[1 + \alpha \times (75 - 20)\right]$$

$$R_{L0} = 0.683\Omega$$

d.1 Calculation for metering

a) As per IEC 60044-1 for CTs with accuracy 1.0%, as in subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy with the specified limits.

$I_n = 1A$

$P_{L0} = I_n^2 \times R_{L0}$ (Lead burden)

$P_{L0} = 0.683W$

$P_M = 2.77V \times A$ (Instruments + relay burden)

$P_B = P_M + P_{L0}$ (Total burden on CT)

$P_B = 3.453W$

$P_N = 15V \times A$ (Rated burden of CT)

$\%P = \frac{P_B}{P_N} \times 100$ (% of connected burden of CT)

$\%P = 23.02$

As the connected burden is less than 25% of the rated burden, the accuracy of the CT cannot be maintained within specific limits. Introduction IPCT may resolve the problem.

IPCT Data

CT Ratio	1/1
CT Class	1.0Fs5, 0.75VA
CT Secondary Resistance	1.32Ω
I_{th} for 1 sec.	50A

CT Lead Data
From IPCT to meters in Control Panel

CT Lead size	$2.5mm^2$
CT Lead length	10m (one way distance while considering IPCT in control panel)
CT Resistance/m	0.00741Ω/m at 20°C

CT lead resistance at 75°C =20m (two way length)

$R_{L1} = R_{20} \times L_1 \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00741 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_1 = 20m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L1} = R_{20} \times L_1 \times \left[1 + \alpha \times (75 - 20)\right]$

$R_{L1} = 0.183\Omega$

$I_{n1} = 1A$ (IPCT secondary current)

$P_{L1} = I_{n1}{}^2 \times R_{L1}$ (Lead burden from IPCT to meters in Control Panel)

$P_{L1} = 0.183W$

$P_M = 0.01 \times A$ (Meter burden)

$P_{BC1} = P_M + P_{L1}$ (Total burden on IPCT)

$P_{BC1} = 0.193W$

$P_{BN1} = 0.75V \times A$ (Rated burden of IPCT)

As per IEC 60044-1 for CTs with accuracy 1.0, as in subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy with the specified limits.

$\%P = \frac{P_{BC1}}{P_{BN1}} \times 100$ (% of connected burden of CT)

$\%P = 25.728$

As the connected burden is more than 25% of the rated burden of IPCT, the accuracy of IPCT can be maintained within specific limits.

d.2.1 To verify that the instruments shall be safe under fault condition:

Operating instrument security factor (K_{0FS}) < Rated instrument security factor (K_{RFS} of instrument)

$R_{CT1} = 1.32\Omega$ (IPCT secondary burden)

$P_{I1} = I_{n1}^{2} \times R_{CT1}$ (Internal burden of IPCT)

$P_{I1} = 1.32W$

$P_{BC1} = 0.193W$ (Total burden on IPCT)

With 25% margin

$P_{BC1} = 125\% \times P_{BC1}$

$P_{BC1} = 0.241W$

$K_{nFs1} = 5$ (Nominal IPCT instrument security factor)

$K_{oFs1} = K_{nFs1} \times \frac{P_{BN1} + P_{I1}}{P_{I1} + P_{BC1}}$ (Operating accuracy limit factor)

$K_{oFs1} = 8.587$

$K_{RFs1} = \frac{I_{sc1}}{I_{n1}}$ (Rated instrument security factor of instruments)

$I_{sc1} = 120A$ (Short time current rating of instruments)

$I_{n1} = 1A$ (Nominal current rating of instrument)

$K_{RFs1} = \frac{I_{sc1}}{I_{n1}}$

$K_{RFs1} = 120$

As $K_{oFs1} < K_{RFs1}$

The available operating instrument security factor is less than rated instrument security factor of instruments. Hence the proposed IPCT data is suitable for required metering application.

d.2.2 To verify that the instruments shall be safe under fault condition:

The thermal rating of the instruments shall be comparable with the thermal rating of the IPCT.

The instruments i.e. Power Measurement make are having the following short time rating:

120 x In /Is

The thermal rating of the instrument connected:

Im = 120 x In1	(Overload rating of the meter in amperes)
tm = 1	(Time period in second for which the meters can be overloaded)
$Km = Im^2 \times tm$	(Thermal rating of the meters)

Thermal rating of the IPCT at operating instrument security factor:

$I_{IPCT} = K_{0Fs1} \times In1$	(IPCT current in ampere at operating instrument security factor)
$t_{IPCT} = 1$	(IPCT thermal withstand time in second)
$K_{IPCT} = I_{IPCT}^2 \times t_{IPCT}$	(Thermal rating of IPCT at operating instrument limit factor)

At IPCT operating instrument security factor, the meter should withstand the fault current for the time specified below:

$t_{mR} = \frac{I_{IPCT}^2 \times t_{IPCT}}{I_m^2}$	(Required withstand time of meters under fault conditions)
$t_{mR} = 3.052 \times 10^{-3}$	(Required withstand time of meters in second)

The factor of safety 'FoS' may be calculated as follows:

$K_{mR} = I_m^2 \times t_{mR}$	(Required thermal rating of the meter)

$$FoS = \frac{K_m}{K_{mR}}$$

FoS = 327.641

The required time for instruments to withstand against the fault is less than the withstand capability of the instruments. Hence the instruments are suitable for required metering application.

d.2.3 To verify that IPCTs shall be safe under fault condition against main CT:

$P_{BC2} = P_{BC1r} + P_{L2} + P_{tP}$ (Total burden on main CT)

$P_{BC1r} = \frac{P_{BC1} + P_{ts}}{n^2}$ (Reflected burden of IPCT, its circuit and relays)

$n = \frac{1}{1}$ (IPCT ratio)

$P_{ts} = I_{n1}{}^2 \times R_{CT1}$

$P_{ts} = 1.32$ W

$P_{BC1r} = \frac{P_{BC1} + P_{ts}}{n^2}$ (Total burden on IPCT)

$P_{BC1r} = 1.561$ W

CT Lead Data
From main CT to control panel

CT Lead size	$4mm^2$
CT Lead length	55m (one way distance while considering IPCT in control panel)
CT Resistance/m	0.00461Ω/m at 20°C
CT lead resistance at 75°C	110m (two way length)

$R_{L2} = R_{20} \times L_2 \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_2 = 110m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L2} = R_{20} \times L_2 \times [1 + \alpha \times (75 - 20)]$

$R_{L2} = 0.626\Omega$

$I_{n2} = 1A$ (Main CT secondary current)

$P_{L2} = I_{n2}{}^2 \times R_{L2}$ (Lead burden from meters to transducers)

$P_{L2} = 0.626W$

$P_R = 2.76V \times A$ (Relay burden)

$R_{CT11} = 1.32\Omega$ (IPCT primary side resistance)

$P_{tP} = I_{n2}^{\ 2} \times R_{CT11}$

$P_{tP} = 1.32W$

$P_{BC2} = P_{BC1r} + P_{L2} + P_{tP} + P_R$

$P_{BC2} = 6.267W$

$P_{BC2} = 125\% \times P_{BC2}$

$P_{BC2} = 7.834W$

$R_{CT2} = 5\Omega$ (Main CT secondary resistance)

$P_{I2} = I_{n2}^{\ 2} \times R_{CT2}$

$P_{I2} = 5W$

$P_{BN2} = 15V \times A$ (Main CT rated burden)

$K_{nFs2} = 20$ (Nominal main CT instrument security factor)

$K_{oFs2} = K_{nFs2} \times \left(\frac{P_{BN2} + P_{I2}}{P_{I2} + P_{BC2}} \right)$ (Operating accuracy limit factor)

$K_{oFs2} = 31.167$

For the safety of IPCT during fault condition, the thermal rating of the IPCT shall be comparable with the thermal rating of the main CT.

The IPCT proposed short time rating:

50A for 1 second

The thermal rating of the IPCT:

$I_{IPCT} = 50$ (Thermal rating of the IPCT in amperes)

$t_{IPCT} = 1$ (Time period in second for which the IPCT can be overloaded without suffering harmful effect)

$K_{IPCT} = I_{IPCT}^{2} \times t_{IPCT}$ (Thermal rating of the IPCT)

The main CT is having the following short time rating:

The thermal rating of the main CT:

$I_{n2} = 1$ (Main CT secondary current in amperes)

$I_{th} = K_{oFs2} \times I_{n2}$ (Primary thermal rating of the CT in amperes)

$t_{CT} = 1$ (Time period in second for which the CT can be overloaded without suffering harmful effect)

$K_{CT} = I_{th}^{2} \times t_{CT}$ (Thermal rating of the CT)

At main CT thermal overloading, the IPCT should withstand the fault current for the time specified below:

$t_{IPCTR} = \frac{I_{th}^{2} \times t_{CT}}{I_{IPCT}^{2}}$ (Required withstand time of meters under fault conditions)

$t_{IPCTR} = 0.389$ (Required withstand time of IPCT in seconds)

The factor of safety 'FoS' may be calculated as follows:

$K_{IPCTR} = I_{IPCT}^{2} \times t_{IPCTR}$ (Required thermal rating of the meter)

$$FoS = \frac{K_{IPCT}}{K_{IPCTR}}$$

$FoS = 2.574$

The required time for IPCT to withstand against the fault is less than the withstand capability of the IPCT. Hence the IPCTs are suitable for required metering application.

<u>Calculation for Protection Relays:</u>

To ensure CT8-1 does not saturate under fault condition:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{RALF})

$P_{L2} = 0.626W$ (CT lead resistance for lead from main CT to IPCT)

$R_{CT} = 5\Omega$

$P_{I} = I_{n}^{2} \times R_{CT}$ (Internal burden on CT)

$P_I = 5W$

$P_B = P_R + P_{L2} + P_{BC1r} + P_{tp}$ (Total burden on CT)

$P_R = 2.76 \times A$ (Total relay burden)

$P_B = P_R + P_{L2} + P_{BC1r} + P_{tp}$

$P_B = 6.267W$

With 25% margin

$P_{B1} = 1.25 \times P_B$

$P_{B1} = 7.834W$

To ensure CT does not saturate:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{RALF})

$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$ (Operating accuracy limit factor)

$K_{ALF} = 20$ (CT accuracy limit factor)

$P_N = 15V \times A$

$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$

$K_{OALF} = 31.167$

$K_{ALFR} = \frac{I_{SC}}{I_P}$

$I_{SC} = 50 \times 10^3 A$

$I_P = 4000A$

$K_{ALFR} = \frac{I_{SC}}{I_P}$

$K_{ALFR} = 12.5$

As $K_{OALF} > K_{ALFR}$

The available operating accuracy limit factor is more than required accuracy limit factor. Hence the proposed CT8-1 data is suitable for required protection application.

Note: CT lead size shall be $4mm^2$ for CT8-1

e. 380KV GIS CTs for 380kV Bus Section

e.1. Device to be connected and its data:

- Digital multimeter power measurement type 6200 :0.01VA
- 50BF (1) Relay ABB Type RAICA :0.02VA
- 50BF (2) Relay Areva Type MCT140 :2.64VA

Total Burden :2.67VA
Overload rating for all metering instruments :120A x I_n

CT Data

CT Ratio	4000/1
CT Class	5P20/1.0, 15VA
CT Secondary Resistance	5Ω

CT Lead Data

CT Lead size	$4mm^2$
CT Lead length	60m (one way distance as a worst case for K130 has been taken)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2× 60 =120m (two way)

$$R_{L0} = R_{20} \times L_0 \times \left[1 + \alpha \times (75 - 20)\right]$$ (CT lead resistance at 75°C)

$$R_{20} = 0.00461 \frac{\Omega}{m}$$ (Resistivity of copper conductor of CT lead)

$$L_0 = 120m$$ (CT lead length)

$$\alpha = \frac{1}{234.5}$$ (Co-efficient of thermal expansion of copper conductor)

$$R_{L0} = R_{20} \times L_0 \times \left[1 + \alpha \times (75 - 20)\right]$$

$$R_{L0} = 0.683\Omega$$

e.2. To verify that accuracy of the CT shall be maintained:

As per IEC 60044-1 for CTs with accuracy 1.0, as in subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy with the specified limits.

$I_n = 1A$

$P_{L0} = I_n^2 \times R_{L0}$ (Lead burden)

$P_{L0} = 0.683W$

$P_M = 2.67V \times A$ (Instrument + relay burden)

$P_B = P_M + P_{L0}$ (Total burden on CT)

$P_B = 3.353W$

$P_N = 15V \times A$ (Rated burden of CT)

$\%P = \frac{P_B}{P_N} \times 100$ (% of connected burden of CT)

$\%P = 22.353$

As the connected burden is more than 25% of the rated burden, the accuracy of the CT cannot be maintained within specified limits.
IPCT Data

CT Ratio	1/1
CT Class	1.0Fs5, 0.75VA
CT Secondary Resistance	1.32Ω
I_{th} for 1 sec.	50A

CT Lead Data
From IPCT to meters in Control Panel

CT Lead size	$2.5mm^2$
CT Lead length	10m (one way distance while considering IPCT in control panel)
CT Resistance/m	0.00741Ω/m at 20°C

CT lead resistance at 75°C =20m (two way length)

$R_{L1} = R_{20} \times L_1 \times \left[1 + \alpha \times (75 - 20)\right]$ (CT lead resistance at 75°C)

$R_{20} = 0.00741 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_1 = 20m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L1} = R_{20} \times L_1 \times \left[1 + \alpha \times (75 - 20)\right]$

$R_{L1} = 0.183\Omega$

$I_{n1} = 1A$ (IPCT secondary current)

$P_{L1} = I_{n1}^{2} \times R_{L1}$ (Lead burden from IPCT to meters in Control Panel)

$P_{L1} = 0.183W$

$P_M = 0.01 \times A$ (Meter burden)

$P_{BC1} = P_M + P_{L1}$ (Total burden on IPCT)

$P_{BC1} = 0.193W$

$P_{BN1} = 0.75V \times A$ (Rated burden of IPCT)

As per IEC 60044-1 for CTs with accuracy 1.0, as in subject case, the connected burden of CT shall be between 25% to 100% of the rated burden to maintain accuracy with the specified limits.

$\%P = \frac{P_{BC1}}{P_{BN1}} \times 100$ (% of connected burden of CT)

$\%P = 25.728$

As the connected burden is more than 25% of the rated burden of IPCT, the accuracy of IPCT can be maintained within specific limits.

e.2.1. To verify that the instruments shall be safe under fault condition:

Operating instrument security factor (K_{0FS}) < Rated instrument security factor (K_{RFS} of instrument)

$R_{CT1} = 1.32\Omega$ (IPCT secondary burden)

$P_{I1} = I_{n1}^{2} \times R_{CT1}$ (Internal burden of IPCT)

$P_{I1} = 1.32W$

$P_{BC1} = 0.193W$ (Total burden on IPCT)

With 25% margin

$P_{BC1} = 125\% \times P_{BC1}$

$P_{BC1} = 0.241W$

$K_{nFs1} = 5$ (Nominal IPCT instrument security factor)

$K_{oFs1} = K_{nFs1} \times \frac{P_{BN1} + P_{I1}}{P_{I1} + P_{BC1}}$ (Operating accuracy limit factor)

$K_{oFs1} = 6.63$

$K_{RFs1} = \frac{I_{sc1}}{I_{n1}}$ (Rated instrument security factor of instruments)

$I_{sc1} = 120A$ (Short time current rating of instruments)

$I_{n1} = 1A$ (Nominal current rating of instrument)

$K_{RFs1} = \frac{I_{sc1}}{I_{n1}}$

$K_{RFs1} = 120$

As $K_{oFs1} < K_{RFs1}$

The available operating instrument security factor is less than rated instrument security factor of instruments. Hence the proposed IPCT data is suitable for required metering application.

e.2.2 To verify that the instruments shall be safe under fault condition:

The thermal rating of the instruments shall be comparable with the thermal rating of the IPCT.

The instruments i.e. Power Measurement make and CEWE make are having the following short time rating:

120 x In /Is

The thermal rating of the instrument connected:

Im = 120 x In1	(Overload rating of the meter in amperes)
tm = 1	(Time period in second for which the meters can be overloaded)
$Km = Im^2 \times tm$	(Thermal rating of the meters)

Thermal rating of the IPCT at operating instrument security factor:

$I_{IPCT} = K_{0Fs1} \times In1$	(IPCT current in ampere at operating instrument security factor)
$t_{IPCT} = 1$	(IPCT thermal withstand time in second)
$K_{IPCT} = I_{IPCT}^2 \times t_{IPCT}$	(Thermal rating of IPCT at operating instrument limit factor)

At IPCT operating instrument security factor, the meter should withstand the fault current for the time specified below:

$$t_{mR} = \frac{I_{IPCT}^2 \times t_{IPCT}}{I_m^2}$$

(Required withstand time of meters under fault conditions)

$t_{mR} = 3.052 \times 10^{-3}$ (Required withstand time of meters in second)

The factor of safety 'FoS' may be calculated as follows:

$$K_{mR} = I_m^2 \times t_{mR}$$

(Required thermal rating of the meter)

$$FoS = \frac{K_m}{K_{mR}}$$

FoS = 327.641

The required time for instruments to withstand against the fault is less than the withstand capability of the instruments. Hence the instruments are suitable for required metering application.

e.2.3. To verify that IPCTs shall be safe under fault condition against main CT:

$P_{BC2} = P_{BC1r} + P_{L2} + P_{tP}$ (Total burden on main CT)

$P_{BC1r} = \frac{P_{BC1} + P_{ts}}{n^2}$ (Reflected burden of IPCT, its circuit and relays)

$n = \frac{1}{1}$ (IPCT ratio)

$P_{ts} = I_{n1}{}^2 \times R_{CT1}$

$P_{ts} = 1.32$ W

$P_{BC1r} = \frac{P_{BC1} + P_{ts}}{n^2}$ (Total burden on IPCT)

$P_{BC1r} = 1.561$ W

CT Lead Data
From main CT to control panel

CT Lead size	$4mm^2$
CT Lead length	55m (one way distance while considering IPCT in control panel)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C 110m (two way length)

$R_{L2} = R_{20} \times L_2 \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L_2 = 110m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_{L2} = R_{20} \times L_2 \times [1 + \alpha \times (75 - 20)]$

$R_{L2} = 0.626\Omega$

$I_{n2} = 1A$ (Main CT secondary current)

$P_{L2} = I_{n2}{}^2 \times R_{L2}$ (Lead burden from meters to transducers)

$P_{L2} = 0.626W$

$P_R = 2.66V \times A$ (Relay burden)

$R_{CT11} = 1.32\Omega$ (IPCT primary side resistance)

$P_{tP} = I_{n2}{}^2 \times R_{CT11}$

$P_{tP} = 1.32W$

$P_{BC2} = P_{BC1r} + P_{L2} + P_{tP} + P_R$

$P_{BC2} = 6.167W$

$P_{BC2} = 125\% \times P_{BC2}$

$P_{BC2} = 7.709W$

$R_{CT2} = 5\Omega$ (Main CT secondary resistance)

$P_{I2} = I_{n2}{}^2 \times R_{CT2}$

$P_{I2} = 5W$

$P_{BN2} = 15V \times A$ (Main CT rated burden)

$K_{nFs2} = 20$ (Nominal main CT instrument security factor)

$K_{oFs2} = K_{nFs2} \times \left(\frac{P_{BN2} + P_{I2}}{P_{I2} + P_{BC2}} \right)$ (Operating accuracy limit factor)

$K_{oFs2} = 31.474$

For the safety of IPCT during fault condition, the thermal rating of the IPCT shall be comparable with the thermal rating of the main CT.

The IPCT proposed short time rating:

50A for 1 second

The thermal rating of the IPCT:

$I_{IPCT} = 50$ (Thermal rating of the IPCT in amperes)

$t_{IPCT} = 1$ (Time period in second for which the IPCT can be overloaded without suffering harmful effect)

$K_{IPCT} = I_{IPCT}^{2} \times t_{IPCT}$ (Thermal rating of the IPCT)

The main CT is having the following short time rating:

The thermal rating of the main CT:

$I_{n2} = 1$ (Main CT secondary current in amperes)

$I_{th} = K_{oFs2} \times I_{n2}$ (Primary thermal rating of the CT in amperes)

$t_{CT} = 1$ (Time period in second for which the CT can be overloaded without suffering harmful effect)

$K_{CT} = I_{th}^{2} \times t_{CT}$ (Thermal rating of the CT)

At main CT thermal overloading, the IPCT should withstand the fault current for the time specified below:

$t_{IPCTR} = \frac{I_{th}^{2} \times t_{CT}}{I_{IPCT}^{2}}$ (Required withstand time of meters under fault conditions)

$t_{IPCTR} = 0.396$ (Required withstand time of IPCT in seconds)

The factor of safety 'FoS' may be calculated as follows:

$K_{IPCTR} = I_{IPCT}^{2} \times t_{IPCTR}$ (Required thermal rating of the meter)

$$FoS = \frac{K_{IPCT}}{K_{IPCTR}}$$

$FoS = 2.524$

The required time for IPCT to withstand against the fault is less than the withstand capability of the IPCT. Hence the IPCTs are suitable for required metering application.

Calculation for Protection Relays:

To ensure CT12-1 does not saturate under fault condition:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{RALF})

$P_{L2} = 0.626W$ (CT lead resistance for lead from main CT to IPCT)

$R_{CT} = 5\Omega$

$P_I = I_n^2 \times R_{CT}$ (Internal burden on CT)

$P_I = 5W$

$P_B = P_R + P_{L2} + P_{BC1r} + P_{tp}$ (Total burden on CT)

$P_R = 2.66 \times A$ (Total relay burden)

$P_B = P_R + P_{L2} + P_{BC1r} + P_{tp}$

$P_B = 6.167W$

With 25% margin

$P_{B1} = 1.25 \times P_B$

$P_{B1} = 7.709W$

To ensure CT does not saturate:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{RALF})

$$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$$ (Operating accuracy limit factor)

$K_{ALF} = 20$ (CT accuracy limit factor)

$P_N = 15V \times A$

$$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$$

$K_{OALF} = 31.474$

$$K_{ALFR} = \frac{I_{SC}}{I_P}$$

$I_{SC} = 50 \times 10^3 A$

$I_P = 4000A$

$K_{ALFR} = \frac{I_{SC}}{I_P}$

$K_{ALFR} = 12.5$

As $K_{OALF} > K_{ALFR}$

The available operating accuracy limit factor is more than required accuracy limit factor. Hence the proposed CT8-1 data is suitable for required protection application.

Note: CT lead size shall be 4mm^2 for CT12-1

<u>Relay to be connected and their data:</u>

- O/C (50P/51 & 50/51N) Relay ABB Type REJ525: 0.1VA

CT Data

CT Ratio	4000/1
CT Class	5P20, 15VA
CT Secondary Resistance	5Ω

CT Lead Data

CT Lead size	4mm^2
CT Lead length	55m (one way consider for K220 as worst case)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2×60 =110m (two way)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 110m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$

$R_L = 0.626\Omega$

$I_n = 1A$

$P_L = I_n^2 \times R_L$

$P_L = 0.626W$

$R_{CT} = 5\Omega$ (CT secondary winding resistance)

$P_I = I_n^2 \times R_{CT}$ (Internal burden on CT)

$P_I = 5W$

$P_B = P_R + P_L$ (Total burden on CT)

$P_R = 0.1V \times A$ (Total relay burden)

$P_B = P_R + P_L$

$P_B = 0.726W$

With 25% margin

$P_{B1} = 1.25 \times P_B$

$P_{B1} = 0.908W$

To ensure CT does not saturate:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{RALF})

$$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$$ (Operating accuracy limit factor)

$K_{ALF} = 20$ (CT accuracy limit factor)

$P_N = 15V \times A$

$$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$$

$K_{OALF} = 67.71$

$$K_{ALFR} = \frac{I_{SC}}{I_P}$$

$I_{SC} = 50 \times 10^3 A$ (Max. through fault short circuit current)

$I_P = 4000A$ (Rated CT primary current)

$$K_{ALFR} = \frac{I_{SC}}{I_P}$$

$$K_{ALFR} = 12.5$$

As $K_{OALF} > K_{ALFR}$

The available operating accuracy limit factor is more than required accuracy limit factor. Hence the proposed CT13-1 data is suitable for required protection application.

Note: CT lead size shall be $4mm^2$ for CT13-1

f. CT Sizing Calculations for 380kV Busbar Protection

All Bay CTs Cores connected with BBP

f.1. Relay to be connected:

Following Two (2) Single Phase High Impedance Differential Relays (one (1) relay for each phase) shall be connected in parallel with the above mentioned CTs.

- High Impedance Busbar Differential Relay (87B-1) ABB Type RADHD
- High Impedance Busbar Differential Relay (87B-2) Alstom Type MFAC14

PROPOSED CT Data

CT Ratio	4000/1
CT Class & data	$X, V_{KP} \geq 500V, I_m \leq 25mA @ 500V$, $R_{CT} \leq 5\Omega$

CT Lead Data

CT Lead size	$4mm^2$
CT Lead length	100m (one way distance for SGT3 as worst case)
CT Resistance/m	0.00461Ω/m at 20°C

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 400m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times \left[1 + \alpha \times (75 - 20)\right]$

$R_L = 2.276\Omega$

f.1.1. <u>High Impedance Busbar Differential Relay (87B-1) ABB Type RADHD</u>

As per relay manufacturer recommendation the knee point voltage (V_{KP}) requirement for RADHD relay is the following:

$V_{KP} \geq 2 \times I_{K2} \times (R_{CT} + R_L)$

Where

$I_{K2} = \frac{I_K}{n}$ (Maximum through fault current related to the secondary side of CT)

$I_K = 50kA$ (Maximum through fault current related to the primary side of CT)

$n = \frac{4000}{1}$ (CT Ratio)

$I_{K2} = \frac{I_K}{n}$

$I_{K2} = 12.5A$

$R_{CT} = 5\Omega$ (Busbar protection CT secondary winding resistance)

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 181.912V$

(With 20% margin)

$V_{KP1} = 1.2 \times V_{KP}$

$V_{KP1} = 218.295V$

f.1.2. High Impedance Busbar Differential Relay (87B-2) Alstom Type MFAC14

As per relay manufacturer recommendation the knee point voltage (V_{KP}) requirement for MFAC14 relay is the following:

$$V_{KP} \geq 2 \times I_f \times (R_{CT} + R_L)$$

Where

$I_f = \frac{I_F}{n}$ (Maximum through fault current related to the secondary side of CT)

$I_F = 50kA$ (Maximum through fault current related to the primary side of CT)

$n = \frac{4000}{1}$ (CT Ratio)

$$I_f = \frac{I_F}{n}$$

$$I_f = 12.5A$$

$R_{CT} = 5\Omega$ (Busbar protection CT secondary winding resistance)

$$V_{KP} \geq 2 \times I_f \times (R_{CT} + R_L)$$

$$V_{KP} = 181.912V$$

(With 20% margin)

$$V_{KP1} = 1.2 \times V_{KP}$$

$$V_{KP1} = 218.295V$$

$U_S = \frac{V_{KP1}}{2}$ (Relay setting voltage)

$$U_S = 109.147V$$

$U_S = 200V$ (Proposed relay setting voltage)

f.2. Primary pick up current to check sensitivity:

$$I_P = N \times \left[(i_{r1} + i_{r2}) + i_{res12} + (i_{sup1} + i_{sup2}) + \sum i_m) \right]$$

Where

$i_{r1} = 70mA$ (Approximate RADHD relay operating current at setting voltage 'U$_S$')

$i_{r2} = 20mA$ (Approximate MFAC14 relay operating current at setting voltage 'U$_S$')

$i_{res12} = 4mA$ (Current through the non-linear resistors of RADHD and MFAC14 at setting voltage 'U$_S$')

$i_m = 10mA$ (megnetizing current at setting voltage 'U$_S$')

$i_{\sup 1} = 1.5mA$ (RAEDK relay current at max. setting)

$i_{\sup 2} = 0.1mA$ (MVTP31 relay current at max. setting)

$N = \frac{4000}{1}$ (CT Ratio)

$$I_P = N \times \left[(i_{r1} + i_{r2}) + i_{res12} + (i_{\sup 1} + i_{\sup 2}) + \sum i_m) \right]$$

$I_P = 502.4A$ (Total primary pickup current)

It is recommended that the sensitivity of the differential relay shall be 10-20% of the CT rated current at setting voltage.

$I_{CT} = 4000A$

$$\%I = \frac{I_P}{I_{CT}} \times 100$$

$\%I = 12.56$

The primary operate current is within recommended value. However, as per SEC requirement the primary operate current shall be 120% of the full load current at setting voltage.

To increase the primary operate current to 120% of the full load current a variable shunt resistor shall be used.

f.3. Calculations for Shunt Resistor

$$I_P = N \times \left[(i_{r1} + i_{r2}) + i_{res12} + (i_{\sup 1} + i_{\sup 2}) + \sum i_m) \right]$$

$I_P = 502.4A$ (Total primary pickup current)

As per SEC requirement the primary pickup current should be such that with opening of one of the CT connected with busbar protection and loading of that feeder equal to the feeder CT rated current i.e. 3000A (which is approximate to 120% of the full load current of feeder) should not cause tripping of the connection protection.

$I_{op} = 3000A$

The primary current 3000A has been adopted as primary operate current for both differential protections.
As the primary pickup current is less than proposed limit. It can be increased by addition of shunt resistor R_{sh}

$I_{op} = 3000A$

$R_{RADHD} = \frac{U_S}{i_{r1}}$ (RADHD relay impedance)

$i_{r1} = 0.07A$ (RADHD relay current at setting voltage)

$U_S = 200V$ (RADHD relay setting voltage)

$R_{RADHD} = \frac{U_S}{i_{r1}}$

$R_{RADHD} = 2.857 \times 10^3 \Omega$

$R_{MFAC14} = \frac{U_S}{i_{r2}}$ (MFAC14 relay impedance)

$i_{r2} = 0.02A$ (MFAC14 relay current at setting voltage)

$U_S = 200V$ (MFAC14 relay setting voltage)

$R_{MFAC14} = \frac{U_S}{i_{r2}}$

$R_{MFAC14} = 1 \times 10^4 \Omega$

$R_{RAEDK} = \frac{U_{RAEDK}}{i_{\sup 1}}$ (RAEDK relay impedance)

$i_{\sup 1} = 1.5 \times 10^{-3} A$ (RAEDK relay current at setting voltage)

$U_{RAEDK} = 80V$ (RAEDK relay max. setting voltage)

$R_{RAEDK} = \frac{U_{RAEDK}}{i_{\sup 1}}$

$R_{RAEDK} = 5.333 \times 10^4 \Omega$

$$R_{MVTP31} = \frac{U_{MVTP31}}{i_{sup2}}$$ (MVTP13 relay impedance)

$$i_{sup2} = 1 \times 10^{-4} A$$ (MVTP13 relay current at max. setting voltage)

$$U_{MVTP31} = 14V$$ (MVTP13 relay max. setting voltage)

$$R_{MVTP31} = \frac{U_{MVTP31}}{i_{sup2}}$$

$$R_{MVTP31} = 1.4 \times 10^{5} \Omega$$

$$R_{r1} = \frac{R_{RADHD} \times R_{RAEDK}}{R_{RADHD} + R_{RAEDK}}$$ (Impedance of the RADHD and RAEDK circuit)

$$R_{r1} = 2.712 \times 10^{3} \Omega$$

$$R_{r2} = \frac{R_{MFAC14} \times R_{MVTP31}}{R_{MFAC14} + R_{MVTP31}}$$ (Impedance of the MFAC14 & MVTP31 circuit)

$$R_{r2} = 9.333 \times 10^{3} \Omega$$

$$I_{Rr1} = I_p \times \frac{R_{r2}}{R_{r1} + R_{r2}}$$ (The primary current through RADHD and RAEDK circuit)

$$I_{Rr1} = 389.289A$$

$$I_{Rr2} = I_p \times \frac{R_{r1}}{R_{r1} + R_{r2}}$$ (The primary current through MFAC14 and MVTP31 circuit)

$$I_{Rr2} = 113.111A$$

The percentage of primary current flowing through RADHD and RAEDK circuit

$$\%I_{Rr1} = \frac{I_{Rr1}}{I_p} \times 100$$

$$\%I_{Rr1} = 77.486$$

The primary current flowing through RADHD and RAEDK circuit if 3000A has been adopted as primary operate current for both differential protections.

$$I_{Rr21} = \frac{\%I_{Rr2}}{100} \times I_{op}$$

$I_{Rr21} = 675.422A$

As the primary pickup current is less than proposed limit. It can be increased by addition of shunt resistor R_{sh}

The calculations for shunt resistor in RADHD and RAEDK circuit

$I_{Rr11} = I_{Rr1} + n \times I_{sh1}$

$I_{sh1} = \frac{I_{Rr11} - I_{Rr1}}{n}$

$I_{sh1} = 0.484A$

$R_{sh1} = \frac{U_S}{I_{sh1}}$ (Resistance of the shunt resistor in RADHD and RAEDK circuit)

$R_{sh1} = 413.375\Omega$

The calculations for shunt resistor in MFAC14 and MVTP31 circuit

$I_{Rr21} = I_{Rr2} + n \times I_{sh2}$

$I_{sh2} = \frac{I_{Rr21} - I_{Rr2}}{n}$

$I_{sh2} = 0.141A$

$R_{sh2} = \frac{U_S}{I_{sh2}}$ (Resistance of the shunt resistor in MFAC14 and MVTP31 circuit)

$R_{sh2} = 1.423 \times 10^3 \Omega$

<u>The Relay Setting</u>

Setting Type	Range	Calculated Settings	Adopted Settings
Operate Voltage	100-400V	109V	200V

The available knee point voltage is more than required knee point voltage. Hence the BBP CT data is suitable for required protection application.

Note: CT lead size shall be 4mm^2 for BBP CTs.

g. CT Sizing Calculations for 380/132 kV Power Transformer Bushing CTs

CT CORE BCT1, BCT2, BCT3 & BCT4 (In Transformer 380kV bushing)

Relays to be connected: SPARE

CT Data

CT Ratio	3000/1
CT Class & data	$X, V_{KP} \geq 400V, I_m \leq 30mA@400V,$ $R_{CT} \leq 5\Omega$

Relay to be connected: High Impedance LV cable Differential Protection (87CP) ABB Type RADHD.

CT Data

CT Ratio	3000/1
CT Class & data	$X, V_{KP} \geq 400V, I_m \leq 30mA@400V,$ $R_{CT} \leq 5\Omega$

CT Lead Data

CT Lead size	$4mm^2$
CT Lead length	150m (one way distance for considered for 132kV GIS for SGT1 as worst case)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2×150 =300m (two way)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 300m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$

$R_L = 1.707\Omega$

g.1. As per relay manufacturer recommendation the knee point voltage (V_{KP}) requirement for RADHD relay is the following:

$V_{KP} \geq 2 \times I_{K2} \times (R_{CT} + R_L)$

Where

$I_{K2} = \frac{I_K}{n}$ (Maximum through fault current related to the secondary side of CT)

$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n \times u_{kr}}$ (Maximum through fault current related to the primary side of CT)

$S_{rT} = 500 \times 10^6 V \times A$ (Rated transformer power)

$U_n = 132kV$

$u_{kr} = 19\%$ (Rated transformer impedance voltage at principal tap)

$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n \times u_{kr}}$

$I_K = 1.151 \times 10^4 A$

$n = \frac{3000}{1}$ (CT Ratio)

$I_{K2} = \frac{I_K}{n}$

$I_{K2} = 3.837A$

$R_{CT} = 5\Omega$ (CT secondary winding resistance)

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 51.469V$

(With 20% margin)

$V_{KP11} = 120\% \times V_{KP}$

$V_{KP11} = 61.762V$

g.2. As per SEC-CTA recommendation the knee point voltage (V_{KP}) requirement for RADHD relay, considering 16 times the rated current of transformer as fault current, is the following:

$V_{KP} \geq 2 \times I_{K2} \times (R_{CT} + R_L)$

Where

$I_{K2} = \frac{16 I_K}{n}$ (Maximum through fault current related to the secondary side of CT)

$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n}$ (Rated current of transformer related to the primary side of CT

$S_{rT} = 500 \times 10^6 V \times A$ (Rated transformer power)

$U_n = 132 kV$

$I_K = \frac{S_{rT}}{\sqrt{3} \times U_n}$

$I_K = 2.187 \times 10^3 A$

$n = \frac{3000}{1}$ (CT Ratio)

$I_{K2} = \frac{16 I_K}{n}$

$I_{K2} = 11.664 A$

$R_{CT} = 5\Omega$ (BCT3 secondary winding resistance)

$V_{KP} = 2 \times I_{K2} \times (R_{CT} + R_L)$

$V_{KP} = 156.465 V$

(With 20% margin)

$V_{KP1} = 120\% \times V_{KP}$

$V_{KP1} = 187.758 V$

$U_S = \frac{V_{KP11}}{2}$ (Relay setting voltage)

$U_S = 30.881 V$

$U_S = 100 V$ (Proposed relay setting voltage)

g.3. Primary operate current:

$I_P = N \times (i_r + i_{res} + \sum i_m + i_{sup})$ (Relay primary operate current)

Where

$i_r = 70mA$ (Approximate RADHD relay operating current at setting voltage 'U$_S$')

$i_{res} = 0mA$ (Current through the non-linear resistor at setting voltage 'U$_S$')

$i_{sup} = 1.5mA$ (RAEDK relay current at max. setting)

$i_m = 12mA$ (CT megnatizing current at setting voltage 'U$_S$')

$\sum = 2$ (Total number of CTs connected to Differential Protection)

$N = \frac{3000}{1}$ (CT Ratio)

$$I_P = N \times (i_r + i_{res} + \sum \times i_m + i_{sup})$$

$$I_P = 286.5A$$

It is recommended that the sensitivity of the differential relay shall be 10-20% of the CT rated current at setting voltage.

$$I_{CT} = 3000A$$

$$\%I = \frac{I_P}{I_{CT}} \times 100$$

$$\%I = 9.55$$

The primary operate current is within recommended value. However, as per SEC requirement the primary operate current shall be 120% of the full load current at setting voltage.

To increase the primary operate current to 120% of the full load current a variable shunt resistor shall be used.

g.4 Calculations for Shunt Resistor

$$I_P = N \times \left(i_r + i_{res} + \sum \times i_m + i_{sup}\right)$$

$I_P = 286.5A$ (Total primary pickup current)

As per SEC requirement the primary pickup current should be such that with opening of one of the CT connected with connection protection and loading of that feeder equal to the CT rated current i.e. 3000A (which is approximated to 120% of the full load current) should not cause tripping of the connection protection.

$$I_{p1} = 3000A$$

The primary current 3000A has been adopted as primary operate current for both differential protections.

As the primary pickup current is less than proposed limit. It can be increased by addition of shunt resistor R_{sh}

$$I_{OP} = N \times \left(i_r + i_{res} + \sum \times i_m + i_{\sup}\right) + N \times I_{sh}$$

$$I_{sh} = \frac{I_{p1} - N \times \left(i_r + i_{res} + \sum \times i_m + i_{\sup}\right)}{N}$$

$$I_{sh} = 0.904A$$

$$R_{sh} = \frac{U_S}{I_{sh}}$$ (Resistance of the shunt resistor in RADHD circuits)

$$R_{sh} = 110.558\Omega$$

<u>The Relay Setting</u>

Setting Type	Range	Calculated Settings	Adopted Settings
Operate Voltage	100-400V	31V	100V

The available knee point voltage is more than required knee point voltage. Hence the CORE BCT5 (In Transformer 132kV bushing), CORE CT8-1 (In 132kV GIS) data is suitable for required protection application.

Note: CT lead size shall be 4mm^2 for CORE BCT5 (In Transformer 132kV bushing), CORE CT8-1 (In 132kV GIS)

Relay to be connected: SPARE

CT Data

CT Ratio	3000/1
CT Class & data	$X, V_{KP} \geq 400V, I_m \leq 30mA@400V,$ $R_{CT} \leq 5\Omega$

CT CORE BCT11 (In Transformer neutral bushing)

Device to be connected and their data:

- FR Hathaway type IDM-T3 : 0.2VA

CT Ratio 3000/1

CT Class & data $X, V_{KP} \geq 400V, I_m \leq 30mA@400V,$ $R_{CT} \leq 5\Omega$

CT Lead size $4mm^2$

CT Lead length 115m (one way distance for SGT3 as worst case)

CT Resistance/m 0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2 × 115 =230m (two way)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 230m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Coefficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$

$R_L = 1.309\Omega$

Total burden on CT in Ohms:

$R_R = \frac{P_R}{I_n^2}$ (Total device resistance)

$R_R = 0.2V \times A$ (Total burden in VA)

$I_n = 1A$

$R_R = \frac{P_R}{I_n^2}$

$R_R = 0.2\Omega$

$R_{CT} = 5.0\Omega$

Calculation of CT requirements for Instantaneous Function

To ensure that a current of maximum fault level does not give saturation, the rated knee point voltage should satisfy the following requirement:

$$V_{KP} = 2I_{kn} \times (R_{CT} + R_L + R_R)$$

$$I_{kn} = \frac{I_{Kn}}{n}$$ (Maximum neutral through fault current related to the secondary side of CT)

$$I_{Kn} = I_{Kp}$$ (Maximum neutral through fault current related to the primary side of CT)

$$I_{Kp} = \frac{S_{rT}}{\sqrt{3} \times U_p \times u_{kr}}$$ (Maximum HV side through fault current related to the primary side of CT)

$$S_{rT} = 500 \times 10^6 V \times A$$ (Rated transformer power)

$$U_p = 380kV$$

$$u_{kr} = 19\%$$ (Rated transformer impedance voltage at principal tap)

$$I_{Kp} = \frac{S_{rT}}{\sqrt{3} \times U_n \times u_{kr}}$$

$$I_{Kp} = 3.998 \times 10^3 A$$ (Maximum neutral short circuit related to the primary side of CT)

$$I_{kn} = \frac{I_{Kn}}{n}$$

$$n = \frac{3000}{1}$$ (CT Ratio)

$$I_{kn} = \frac{I_{Kn}}{n}$$

$$I_{kn} = 1.333A$$

$$V_{KP} = 2I_{kn} \times (R_{CT} + R_L + R_R)$$

$$V_{KP} = 17.35V$$

(With 20% margin)

$V_{KP1} = 1.2 \times V_{KP}$

$V_{KP1} = 20.82V$

The available knee point voltage is more than required knee point voltage. Hence the CORE BCT11 (In Transformer neutral bushing) data is suitable for required FR application.

CT CORE BCT12 (In 33kV tertiary bushing)

Relay to be connected and their data:

- O/C (51-1) Relay ABB Type REJ525 : 0.1VA

CT Data

CT Ratio	2500-800-100/1
CT Class and data	5P20, 15VA
CT Secondary Resistance	0.25Ω

CT Lead Data

CT Lead size	$10mm^2$
CT Lead length	115m (one way distance for SGT3 as worst case)
CT Resistance/m	0.00183Ω/m at 20°C

CT lead resistance at 75°C for 2×115 =230m (two way)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00183 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 230m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$

$R_L = 0.52\Omega$

$I_n = 1A$

$P_L = I_n^2 \times R_L$

$P_L = 0.52W$

$R_{CT} = 0.25\Omega$

$P_I = I_n^2 \times R_{CT}$ (Internal burden on CT)

$P_I = 0.25W$

$P_B = P_R + P_L$ (Total burden on CT)

$P_R = 0.1V \times A$ (Relay burden)

$P_B = P_R + P_L$

$P_B = 0.62W$

With 25% margin

$P_{B1} = 125\% \times P_B$

$P_{B1} = 0.775W$

To ensure CT does not saturate:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{RALF})

$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$ (Operating accuracy limit factor)

$K_{ALF} = 20$ (CT accuracy limit factor)

$P_N = 15V \times A$

$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$

$K_{OALF} = 297.699$

$K_{ALFR} = \frac{I_{Kt}}{I_p}$

$I_{Kt} = \frac{S_{rT}}{\sqrt{3} \times U_{np} \times u_{kr}}$ (Maximum TV side through fault current related to the primary side of CT)

$S_{rT} = 500 \times 10^6 V \times A$ (Rated transformer power)

$U_{nt} = 33kV$

$u_{kr} = 41.1\%$ (Rated transformer impedance voltage HV/TV at principal tap)

$$I_{Kt} = \frac{S_{rT}}{\sqrt{3} \times U_{np} \times u_{kr}}$$

$$I_{Kt} = 2.128 \times 10^4 A$$

$$I_P = 100A$$

$$K_{ALFR} = \frac{I_{Kt}}{I_P}$$

$$K_{ALFR} = 212.84$$

As $K_{OALF} > K_{ALFR}$

The available operating accuracy limit factor is more than required accuracy limit factor. Hence the proposed BCT12 (In 33kV tertiary bushing) data is suitable for required protection application.

Note: CT lead size shall be $10mm^2$ for BCT12 (In 33kV tertiary bushing)

CT CORE BCT13 (In 33kV tertiary bushing)

<u>Relay to be connected and their data:</u>

- Current Relay (51-1) Areva Type REJ525 : 0.1VA

CT Data

CT Ratio	2500-800-<u>100</u>/1
CT Class and data	5P20, 15VA
CT Secondary Resistance	0.25Ω@100/1

CT Lead Data

CT Lead size	$10mm^2$
CT Lead length	115m (one way distance for SGT3 as worst case)
CT Resistance/m	0.00183Ω/m at 20°C

CT lead resistance at 75°C for 2×115 =230m (two way)

$$R_L = R_{20} \times L \times \left[1 + \alpha \times (75 - 20)\right]$$ (CT lead resistance at 75°C)

$$R_{20} = 0.00183 \frac{\Omega}{m}$$ (Resistivity of copper conductor of CT lead)

$L = 230m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times \left[1 + \alpha \times (75 - 20)\right]$

$R_L = 0.52\Omega$

$I_n = 1A$

$P_L = I_n^2 \times R_L$

$P_L = 0.52W$

$R_{CT} = 0.25\Omega$

$P_I = I_n^2 \times R_{CT}$ (Internal burden on CT)

$P_I = 0.25W$

$P_B = P_R + P_L$ (Total burden on CT)

$P_R = 0.1V \times A$ (Relay burden)

$P_B = P_R + P_L$

$P_B = 0.62W$

With 25% margin

$P_{B1} = 125\% \times P_B$

$P_{B1} = 0.775W$

To ensure CT does not saturate:

Operating accuracy limit factor (K_{OALF})>Required accuracy limit factor (K_{RALF})

$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}}\right)$ (Operating accuracy limit factor)

$K_{ALF} = 20$ (CT accuracy limit factor)

$$P_N = 15V \times A$$

$$K_{OALF} = K_{ALF} \times \left(\frac{P_N + P_I}{P_I + P_{B1}} \right)$$

$$K_{OALF} = 297.699$$

$$K_{ALFR} = \frac{I_{Kt}}{I_P}$$

$$I_{Kt} = 2.128 \times 10^4 A$$

$$I_P = 100A$$

$$K_{ALFR} = \frac{I_{Kt}}{I_P}$$

$$K_{ALFR} = 212.84$$

As $K_{OALF} > K_{ALFR}$

The available operating accuracy limit factor is more than required accuracy limit factor. Hence the proposed BCT13 (In 33kV tertiary bushing) data is suitable for required protection application.

Note: CT lead size shall be $10mm^2$ for BCT13 (In 33kV tertiary bushing)

h. <u>CT Sizing Calculations for 380kV Shunt Reactor Bushing CTs</u>

CT CORE BCT1, BCT2 (In SR 380kV bushing)

Relays to be connected: SPARE

CT Data

CT Ratio	600-300/1
CT Class & data	$X, V_{KP} \geq 600V, I_m \leq 150mA @ 600V,$ $R_{ct} \leq 1.6\Omega$ $X, V_{KP} \geq 300V, I_m \leq 300mA @ 300V,$ $R_{ct} \leq 0.8\Omega$

CT CORE BCT5 (In SR neutral bushing)

Device to be connected and their data:

- FR Hathaway type IDM-T3 : 0.2VA

CT Data

CT Ratio	600-300/1
CT Class & data	$X, V_{KP} \geq 600V, I_m \leq 150mA @ 600V,$ $R_{ct} \leq 1.6\Omega$
	$X, V_{KP} \geq 300V, I_m \leq 300mA @ 300V,$ $R_{ct} \leq 0.8\Omega$
CT Lead Data	
CT Lead size	$4mm^2$
CT Lead length	110m (one way for SR2 as worst case)
CT Resistance/m	0.00461Ω/m at 20°C

CT lead resistance at 75°C for 2×110 =220m (two way)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$ (CT lead resistance at 75°C)

$R_{20} = 0.00461 \frac{\Omega}{m}$ (Resistivity of copper conductor of CT lead)

$L = 220m$ (CT lead length)

$\alpha = \frac{1}{234.5}$ (Co-efficient of thermal expansion of copper conductor)

$R_L = R_{20} \times L \times [1 + \alpha \times (75 - 20)]$

$R_L = 1.252\Omega$

Total burden on CT in Ohms:

$R_R = \frac{P_R}{I_n^2}$ (Total device resistance)

$R_R = 0.2V \times A$ (Total burden in VA)

$I_n = 1A$

$R_R = \frac{P_R}{I_n^2}$

$R_R = 0.2\Omega$

$R_{CT} = 1.6\Omega$

Calculation of CT requirements for Instantaneous Function

To ensure that a current of maximum fault level i.e. 50kA does not give saturation, the rated knee point voltage should satisfy the following requirement:

$V_{KP} = 2I_{kn} \times (R_{CT} + R_L + R_R)$

$$I_{Kn} = 10 \times \frac{S_{rS}}{\sqrt{3} \times U_r}$$

(Maximum through fault current related to the primary side of CT has been considered as inrush current of SR which shall be 10 times the rated current of SR)

$$S_{rS} = 80 \times 10^6 V \times A$$

(SR rating in VAR has been considered as VA)

$$U_r = 380kV$$

$$I_{Kn} = 10 \times \frac{S_{rS}}{\sqrt{3} \times U_r}$$

$$I_{kn} = 1.215 \times 10^3 A$$

$$I_{kn} = \frac{I_{Kn}}{n}$$

(Maximum neutral short circuit related to the secondary side of CT)

$$n = \frac{600}{1}$$

(CT Ratio)

$$I_{kn} = \frac{I_{Kn}}{n}$$

$$I_{kn} = 2.026A$$

$$V_{KP} = 2I_{kn} \times (R_{CT} + R_L + R_R)$$

$$V_{KP} = 12.366V$$

(With 20% margin)

$$V_{KP1} = 1.2 \times V_{KP}$$

$$V_{KP1} = 14.839V$$

The available knee point voltage is more than required knee point voltage. Hence the BCT5 (In SR neutral bushing) data is suitable for required FR application.

Note: CT lead size shall be 4mm^2 for BCT5 (In SR neutral bushing)

9.0 ANNEX 2 - SAMPLE OF PT SIZING CALCULATIONS

Part I. Scope

This section covers sample calculation of Potential Transformer (PT) sizing for the metering and protection requirements of a substation and transmission line systems. The objective of this sample calculation is to present a uniform guideline for sizing a potential transformer.

The values are taken from manufacturer's catalogue for the purposes of arriving at the most realistic values during the computation.

Part II. SAMPLE CALCULATION

Shedgum Substation 380kV GIS Typical Future Line 1 (A) and Line 2 (B)

The following table shows the devices connected to winding 1 and 2 of the CCVT=C5–T6. The type of the line protection as well as the future metering units is not defined yet and therefore assumptions for the values were made.

PT details:

Winding 1:

Accuracy: cl. 0.5 + 3P

Burden: 100VA

Ratio: $\frac{380kV}{\sqrt{3}} \Big/ \frac{110}{\sqrt{3}} - 110 \Big/ \frac{110}{\sqrt{3}} - 110V$

Winding 2:

Accuracy: cl. 0.2 + 3P

Burden: 100VA

Ratio: $\frac{380kV}{\sqrt{3}} \Big/ \frac{110}{\sqrt{3}} - 110 \Big/ \frac{110}{\sqrt{3}} - 110V$

Apparatus	Qty	Manuf.	Type	Remark	Consumption per phase/unit (VA)	Consumption total per phase (VA)
CCVT Winding 1						
Fuse Failure Superv. Unit	3	ABB	RXBA4		2	6
Line Protection set 1	1		Unknown		25	25
Fault Recording Unit	1		Unknown		10	10
Total						**41**
CCVT Winding 2						
Fuse Failure Superv. Unit	3	ABB	RXBA4		2	6
Line Protection set 2	1	-	Unknown		25	25
Sync. Check	2	ABB	SPAU		2.5	5
Voltmeter	1	GEC	V96LI		4	4
Volt Transducer	1	GEC	ISTAT300		1.1	1.1
MW,MVAR Transducer	4	GEC	ISTAT300		2.1	8.4
Metering M1,M2	2	-	Unknown		5	10
Total						**59.5**

The above typical sizing calculations provides evidence that the CCVT's with 100VA for each windings are adequate for the given burden. It should be noted that the actual burden of the selected devices such as the line protection units, fault recorder and metering devices with assumed values should not exceed the total of the assumed burden and the available spare capacity of the CCVT.

Per IEC-60044-2, clause 12.2 and 13.2, the CCVT connected burden shall be within 25% to 100% of the selected rating to maintain accuracy within specified limits.